741 Data Analysis Project

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Introduction

Research was conducted measuring lung function of asthmatics and non-asthmatics, with patients followed for up to 12 surveys. Also included in the research was the smoking status of the individual, including the number of cigarettes they smoked per day at the time of the survey. Additionally, the height, weight, and sex of most participants were included. The following data analysis was conducted to determine the relationship between asthma and lung function over time, while factoring in smoking status and the aforementioned physical attributes. Much of this analysis was conducted using mixed effects models, with subjects as a random effect. This allowed the differences between participants to be accounted for mathematically.

Data Cleaning

Before any analysis can be done, the first step is to reformat the dataset into a long format, with successive observations on the same subject moved into its own row. Corresponding FEV, FVC, smoking status, number of cigarettes per day, age, mean height, and mean weight are included for each time point. Additionally, the variable cncig was converted into a categorical variable based on its quartiles, with NA values being replaced with 0. The same was done for age, sans imputing 0 for missing age values. Lastly, sex was re-coded as a binary categorical variable.

Exploratory Data Analysis

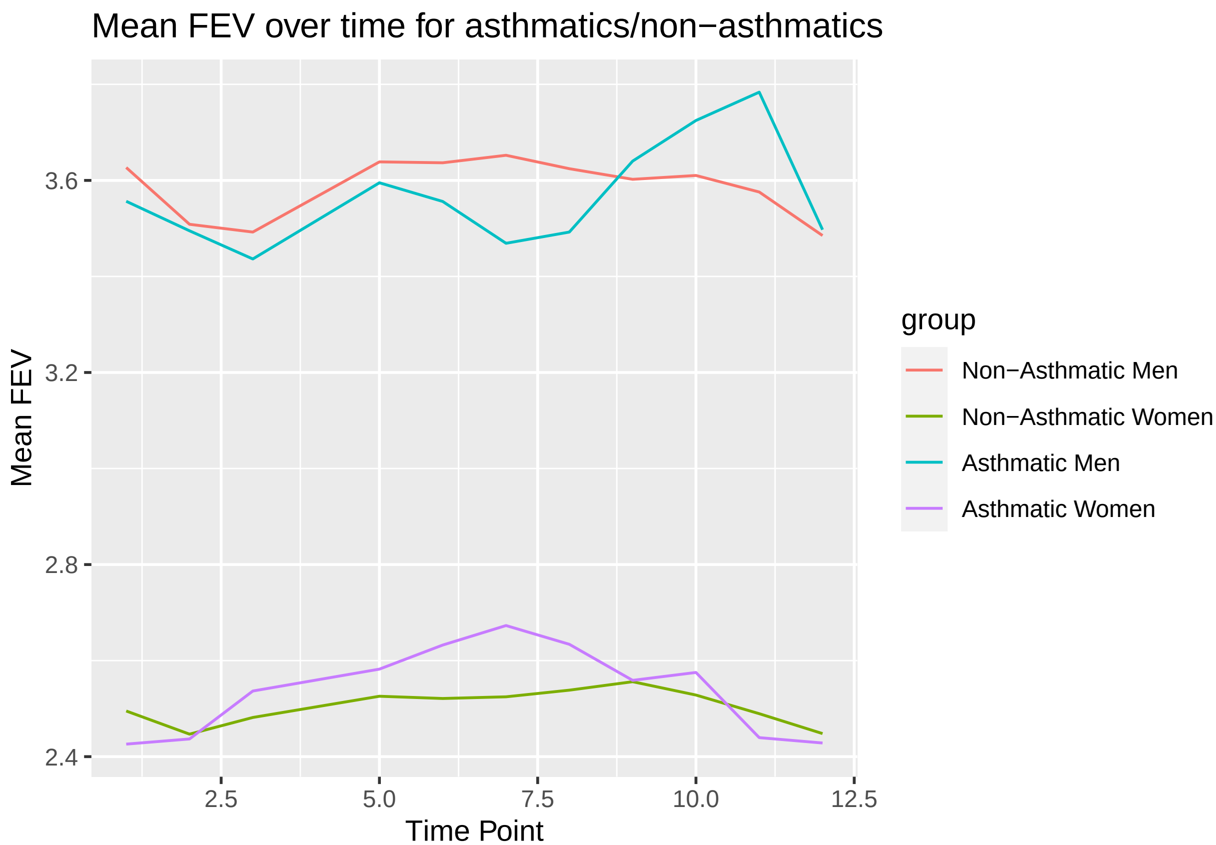
Using the data set which now has no missing values for height or weight, we can begin preliminary data exploration, which may influence model selection and interpretation of the formal analyses. It is known that height and sex can have a significant effect on lung function, potentially over time, so these relationships will be briefly explored. Additionally, it is intuitive that age will have significant effect on lung function, meaning an exploration of this variable will also be useful.

```
##         id             year             fev             cncig
##  1111701:   11   Min.   : 1.000   Min.   :0.535   Min.   : 0.000
##  1112301:   11   1st Qu.: 3.000   1st Qu.:2.241   1st Qu.: 0.000
##  1112901:   11   Median : 6.000   Median :2.912   Median : 0.000
##  1112902:   11   Mean   : 6.067   Mean   :2.973   Mean   : 5.094
##  1113601:   11   3rd Qu.: 9.000   3rd Qu.:3.599   3rd Qu.: 0.000
##  1114501:   11   Max.   :12.000   Max.   :6.104   Max.   :98.000
##  (Other):13904                    NA's   :4057    NA's   :1539
##       smk              fvc             age             sex
##  Min.   :0.0000   Min.   :0.604   Min.   : 20.08   Male  :6168
##  1st Qu.:0.0000   1st Qu.:2.909   1st Qu.: 36.75   Female:7802
##  Median :0.0000   Median :3.672   Median : 52.50
##  Mean   :0.2204   Mean   :3.756   Mean   : 53.01
```

```
## 3rd Qu.:0.0000   3rd Qu.:4.492   3rd Qu.: 68.92
## Max.   :1.0000   Max.   :7.616   Max.   :101.83
## NA's   :324      NA's   :4073    NA's   :18
##     asthma             mht            family          mwt
## Min.   :0.00000   Min.   :51.00   11196  :   48   Min.   : 75.05
## 1st Qu.:0.00000   1st Qu.:63.00   23411  :   48   1st Qu.:131.29
## Median :0.00000   Median :66.00   11359  :   45   Median :150.62
## Mean   :0.08926   Mean   :65.89   11188  :   41   Mean   :154.47
## 3rd Qu.:0.00000   3rd Qu.:69.00   21206  :   41   3rd Qu.:173.10
## Max.   :1.00000   Max.   :77.00   22152  :   41   Max.   :280.07
## NA's   :112       NA's   :3036    (Other):13706   NA's   :38
##   smoke_ever        cncig_quartile      age_quartile
## Min.   :0.0000   Lowest 25% :10132   20.0-36.7  :3481
## 1st Qu.:0.0000   25%-50%    :  370   36.8-52.5  :3489
## Median :0.0000   50%-75%    : 1085   52.6-68.9  :3483
## Mean   :0.3435   Highest 25%:  844   68.91-101.8:3499
## 3rd Qu.:1.0000   NA's       : 1539   NA's       :  18
## Max.   :1.0000
##
```

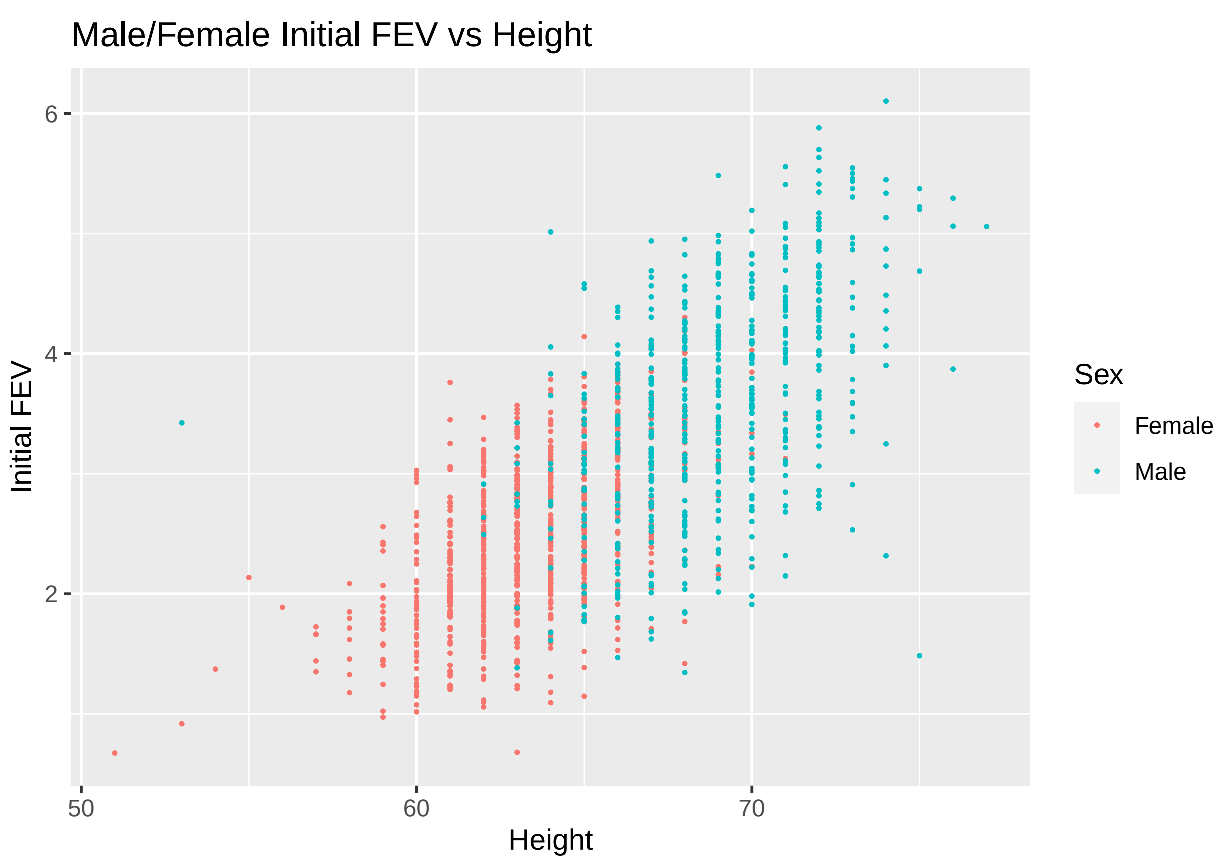
Comments: - Number of cigarettes per day ranges from 0 to 98 - The vast majority of participants are in the lowest 25% of cigarettes smoked per day - Age ranges from 20 to 102 - The quartiles are fairly evenly distributed - Height ranges from 4'3" to 6'5" - Weight ranges from 75 to 280 pounds - There are almost 1700 more females in the study than males, though both sample sizes are large

Gender



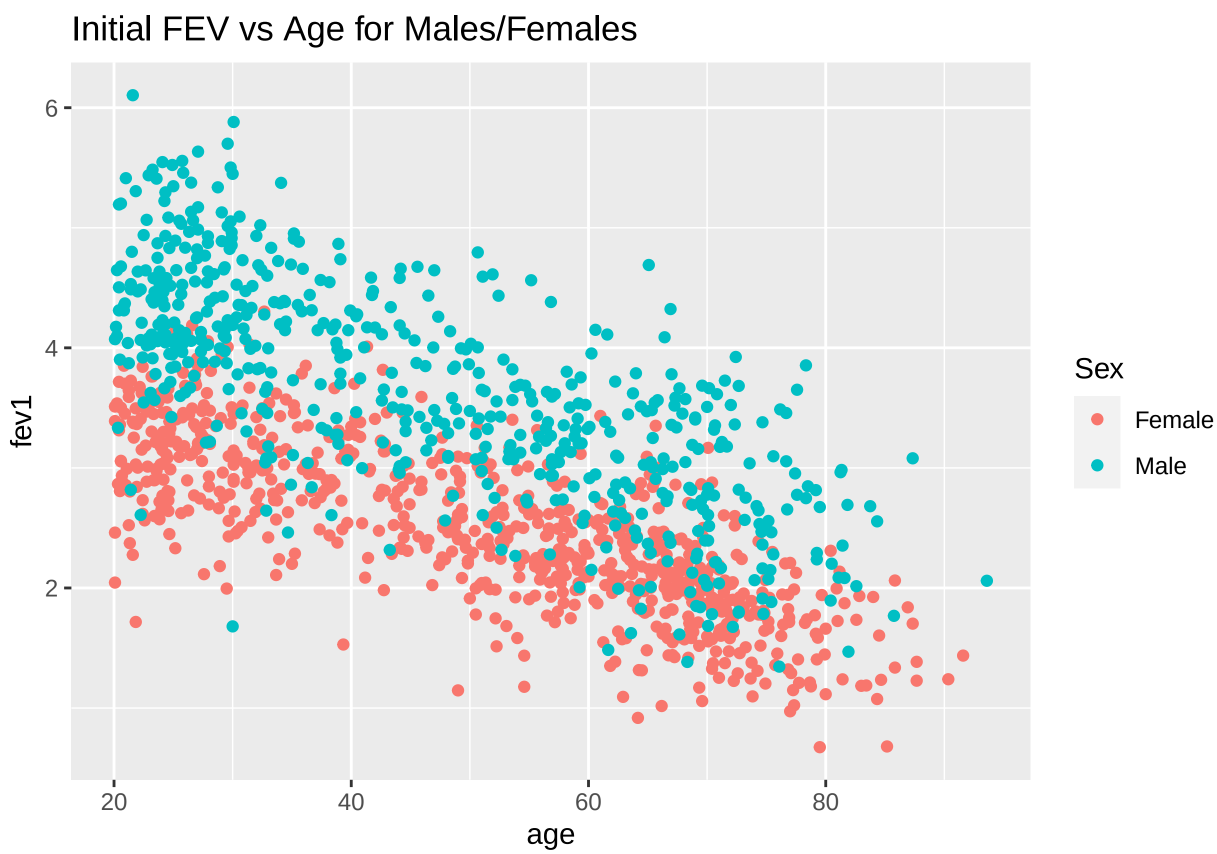
It is clear that males have higher mean FEVs than females, regardless of whether or not they are asthmatic. Further, it appears that being asthmatic does not largely affect the FEV of either males or females over time, with the respective curves being very close to the non-asthmatic curves. Notably, lung function does not appear to decrease significantly over time for any of the four groups, likely suggesting other factors or at play. That is, being male/female or asthmatic/non-asthmatic has little effect on lung function over time. However, this result may be misleading because the mean FEVs were calculated at each time point, ignoring any potential differences between patients. This is where a random effects model will come into play later.

Height *** COMPARE MISSING HEIGHT FEV WITH KNOWN HEIGHT FEV



The box plots indicate a clear difference in heights between men and women, most of the men in the data set being taller than most of the women in the data set. The scatter plot once again shows that men have higher FEVs than women in general, but it a positive linear correlation between FEV and height can be seen. In other words, taller individuals would be expected to have greater lung function. Note that this does not include the time variable, taller individuals may or may not have worse lung function over time than shorter individuals.

Age

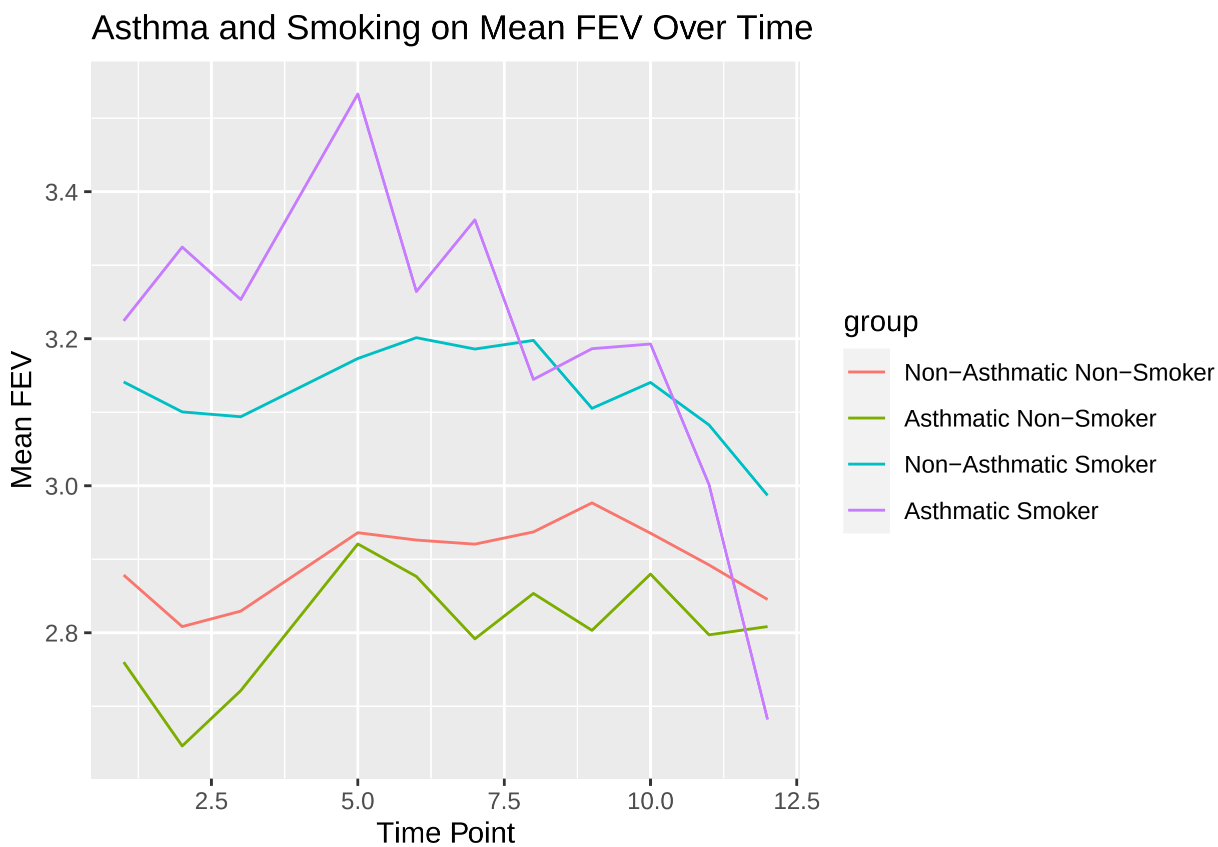


It can be seen for both males and females that the initial FEV of a participant is lower if they are of an older age. This relationship also appears to be linear, and suggests that age is an important factor in predicting lung function independent of asthma and/or smoking history. It should also be noted that the three of these variables are highly dependent upon each other, intuitively. In general, males are taller than females, and there is a chance that people in their twenties can grow in height. It is natural to assume that height, sex, and age are all independent of a participant having asthma. The correlation between explanatory variables could lead to difficulty in model fitting due to multicollinearity, which may reduce the significance of parameter estimates. This must be considered through the rest of the report.

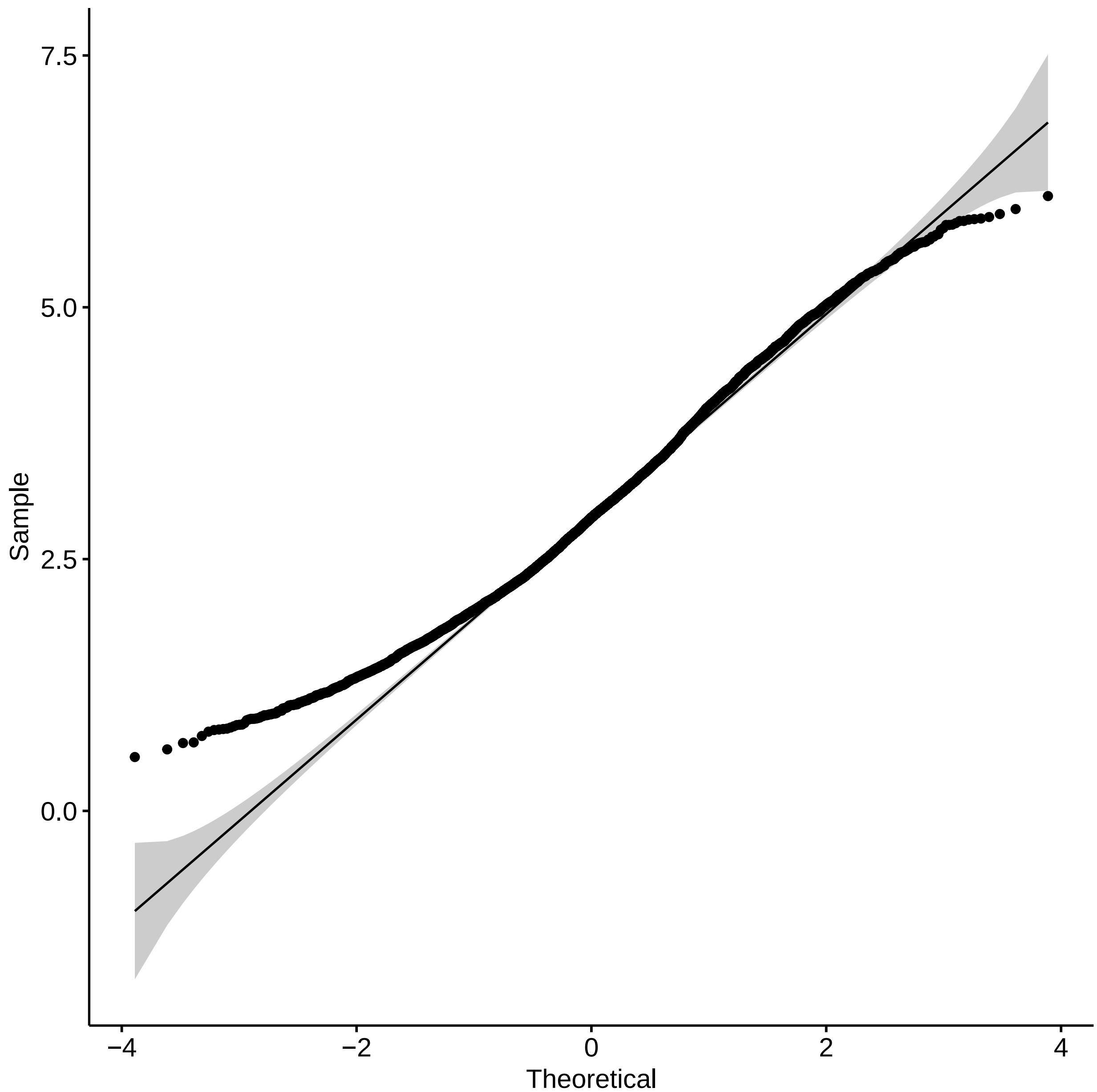
Asthma vs Smoking

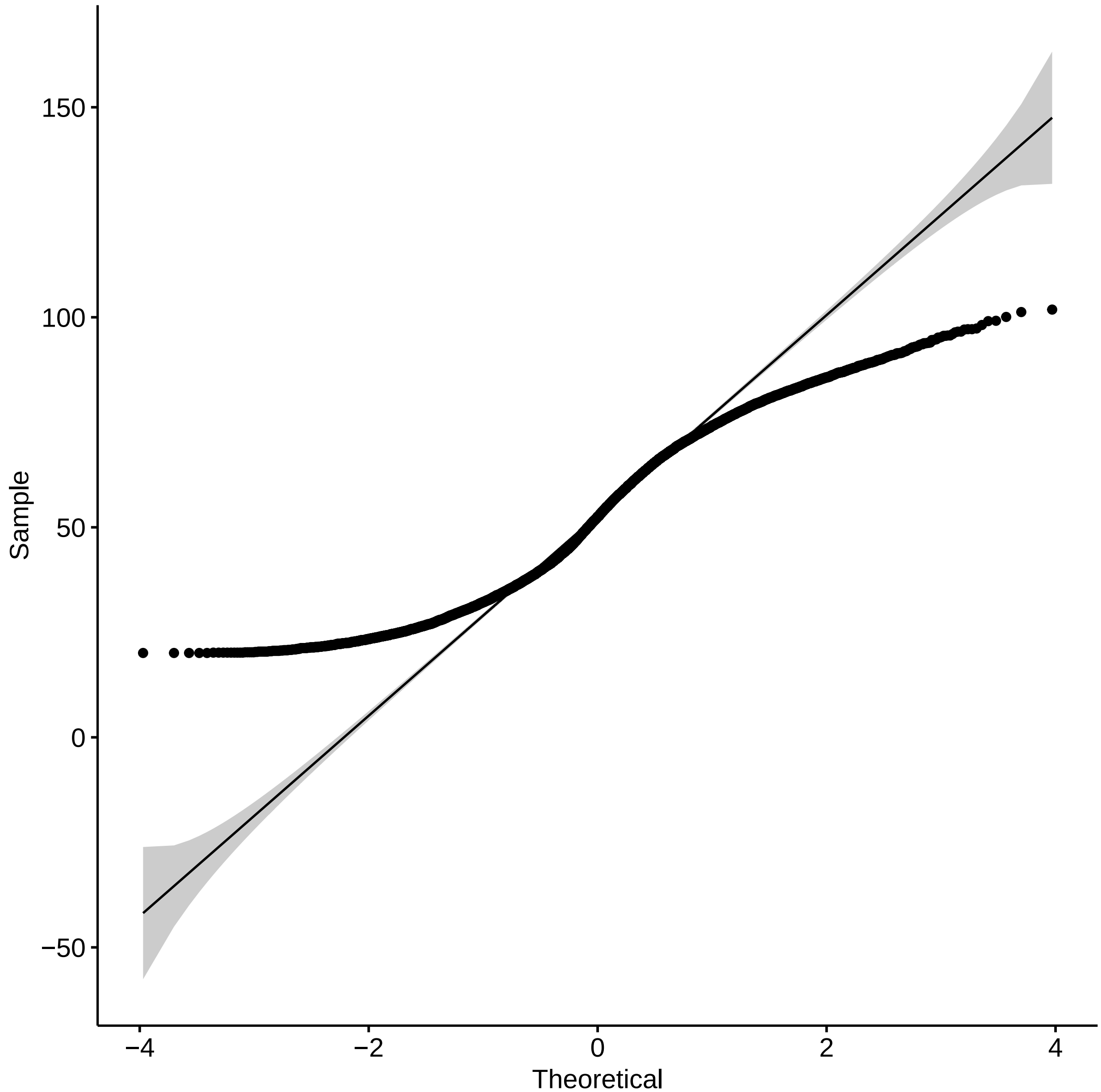
```
as_data <-
  family_asthma_long %>%
  group_by(year, asthma, smoke_ever) %>%
  reframe(fev = mean(fev, na.rm = TRUE)) %>%
```

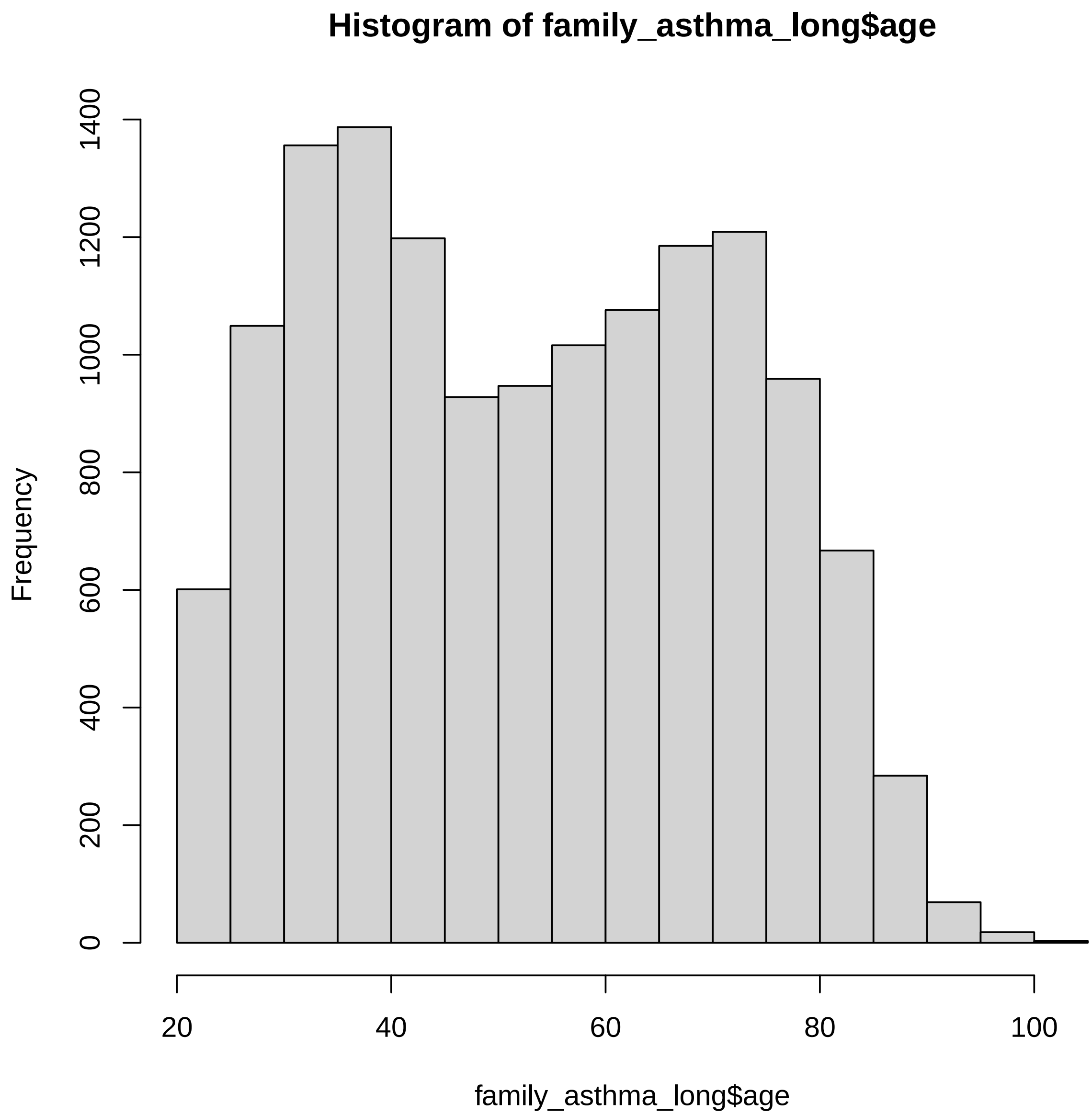
```r
  mutate(group = interaction(asthma,smoke_ever) %>%
           factor(labels =
      c("Non-Asthmatic Non-Smoker", "Asthmatic Non-Smoker",
        "Non-Asthmatic Smoker","Asthmatic Smoker"))) %>%
  filter(!is.na(group))

ggplot(as_data, aes(x=year,y=fev,color=group)) +
  geom_line() +
  ggtitle('Asthma and Smoking on Mean FEV Over Time') +
  labs(x = 'Time Point', y = 'Mean FEV')
```



***COMMENTARY 295 asthmatic smokers, 3063 non-asthmatic smokers, 560 asthmatic non-smokers, 5925 non-asthmatic non-smokers


7.5
5.0
2.5
0.0
Sample
−4
−2
0
2
4
Theoretical


150
100
50
0
−50
Sample
−4
−2
0
2
4
Theoretical




```
##
##  Asymptotic one-sample Kolmogorov-Smirnov test
##
## data:  family_asthma_long$fev
## D = 0.88944, p-value < 2.2e-16
## alternative hypothesis: two-sided
```

We see that age is not normally distributed, rather it is bimodal. However this should not be a problem in the analysis as the age will be analyzed as a categorical variable. FEV was found to non-normally distributed as well, but it should be noted that the significance of this result is greatly affected by the sample size. In larger sample sizes, such as our data set, a small discrepancy from normality will be deemed more significant.

Research Questions

1. Do asthmatics have steeper rates of decline (slope) or lower levels of lung function (FEV1) than non-asthmatics, independent of smoking history?

```r
model_1 =
  family_asthma_long %>%
  lmer(fev ~ year * asthma + # Main Coefficients of Interest
       mht + sex  + age_quartile + # independent controls
       (1 | family / id) + (age | id) , data = .)

regression_table(model_1)
```

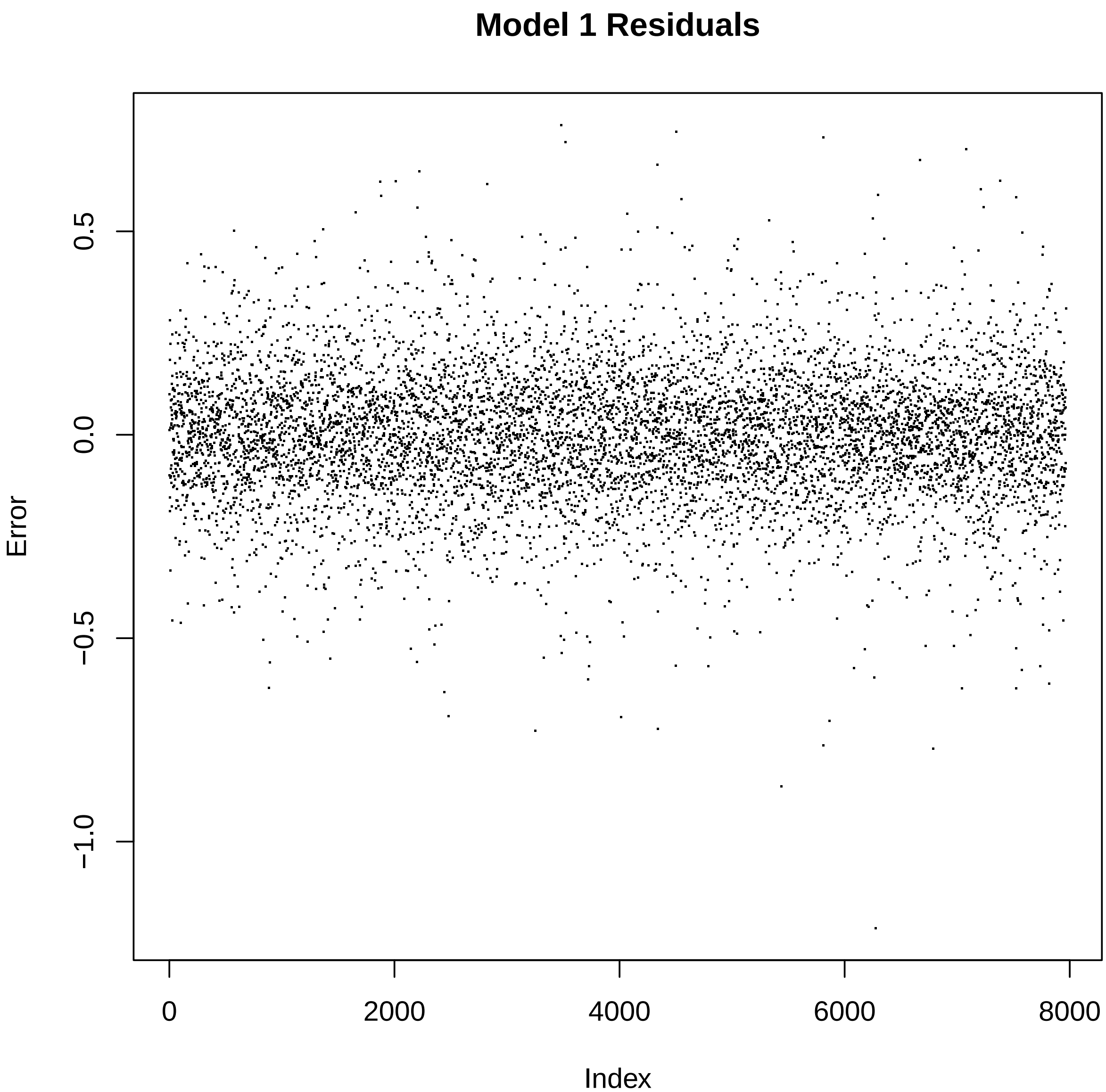
Characteristic	Beta (SE)
Time (years)	-0.018*** (0.001)
Asthmatic	-0.204*** (0.052)
Average Height	0.129*** (0.006)
sex	NA*** (NA)
Male	—
Female	-0.342
Age Quartile	NA*** (NA)
20.0-36.7	—
36.8-52.5	-0.163
52.6-68.9	-0.354
68.91-101.8	-0.465
Time (years) * Asthmatic	0.007 (0.004)

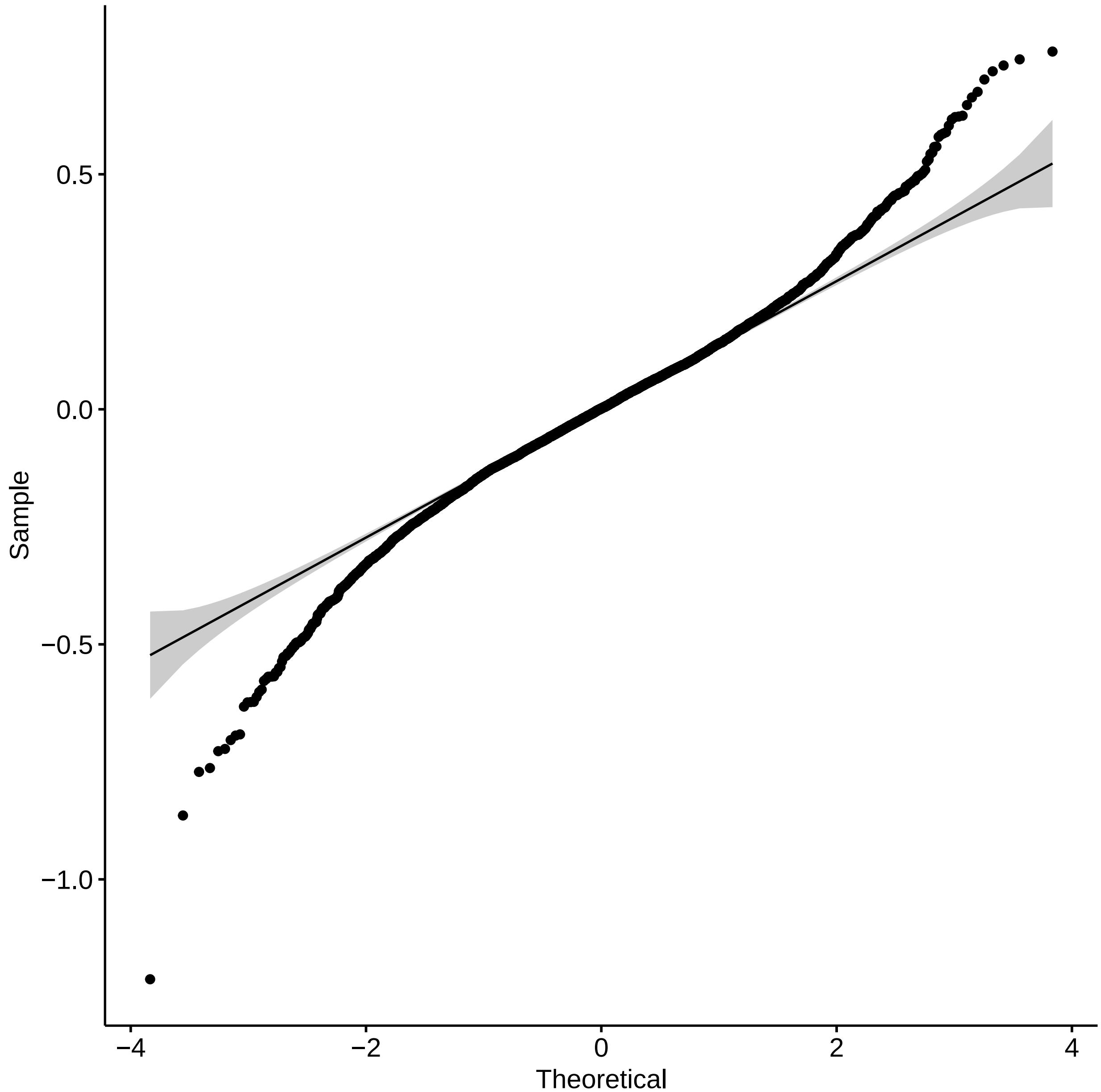
The model predicts FEV based on time, asthma status, the height and sex of the individual, their age quartile, and an interaction between asthma and time. Subjects and subjects by age are treated as random coefficients, the covariance between slopes and intercepts included.
The last parameter estimate being insignificant suggests that asthmatics do not have steeper rates of decline than non-asthmatics (independent of smoking history), but the significant negative estimate for "asthmatic" suggests they do have lower levels of lung function. The negative and significant estimate for "time" indicates that lung function is expected to decrease over time. Additionally, it can be seen that taller individuals have significantly higher levels of lung function, which supports the preliminary data exploration. The EDA also coincides with the result for sex, with females having significantly lower FEV than males. Age was also a significant factor, with the older participants having lower lung function in tiers. That is, the oldest patients had worse lung function than the next oldest, etc. It should be made clear that the only variable interacting with time is "asthma", the above model does not indicate whether differences in height, sex, or age contribute to reduced FEV over time.

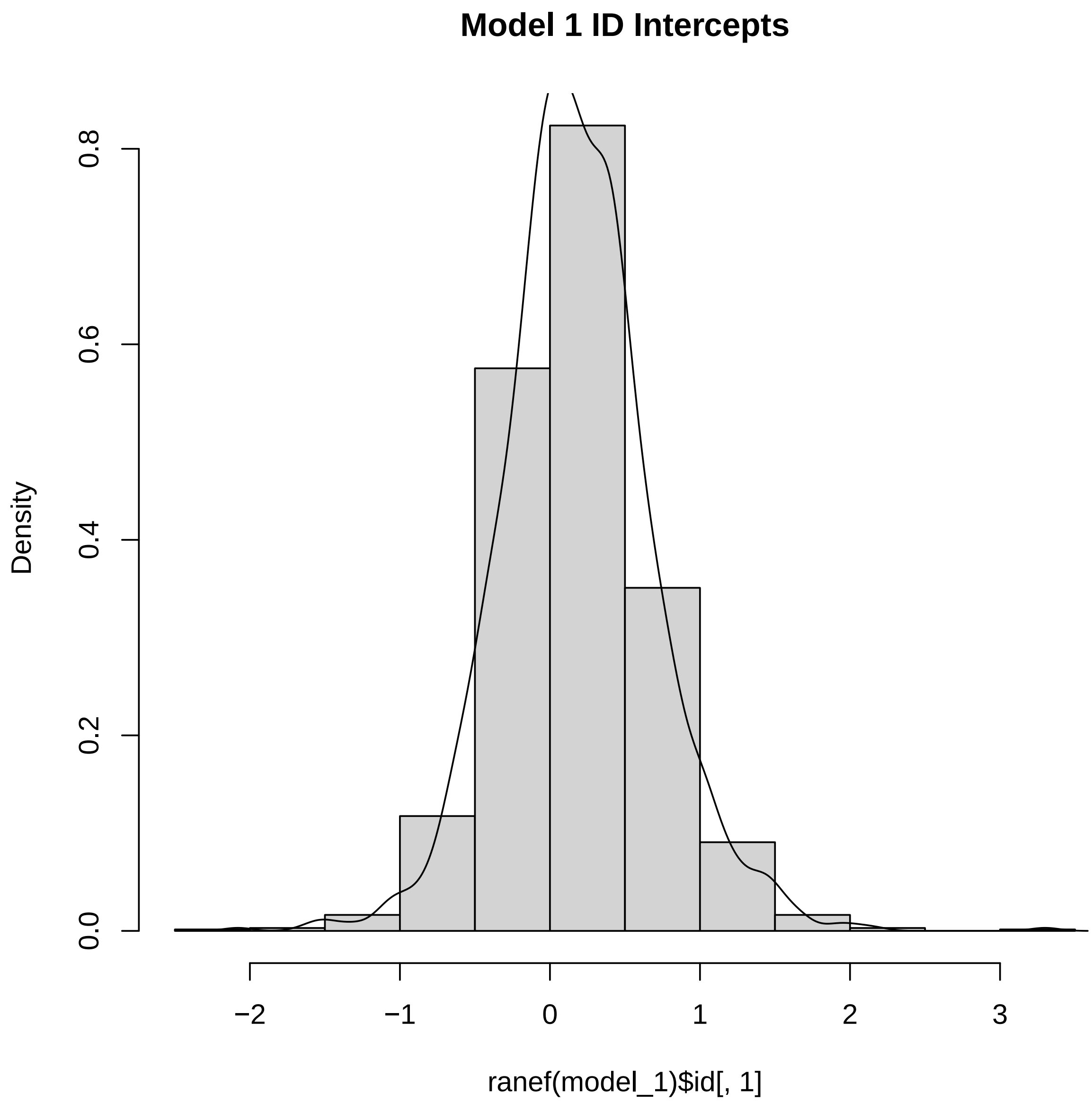
Question 1 Analysis

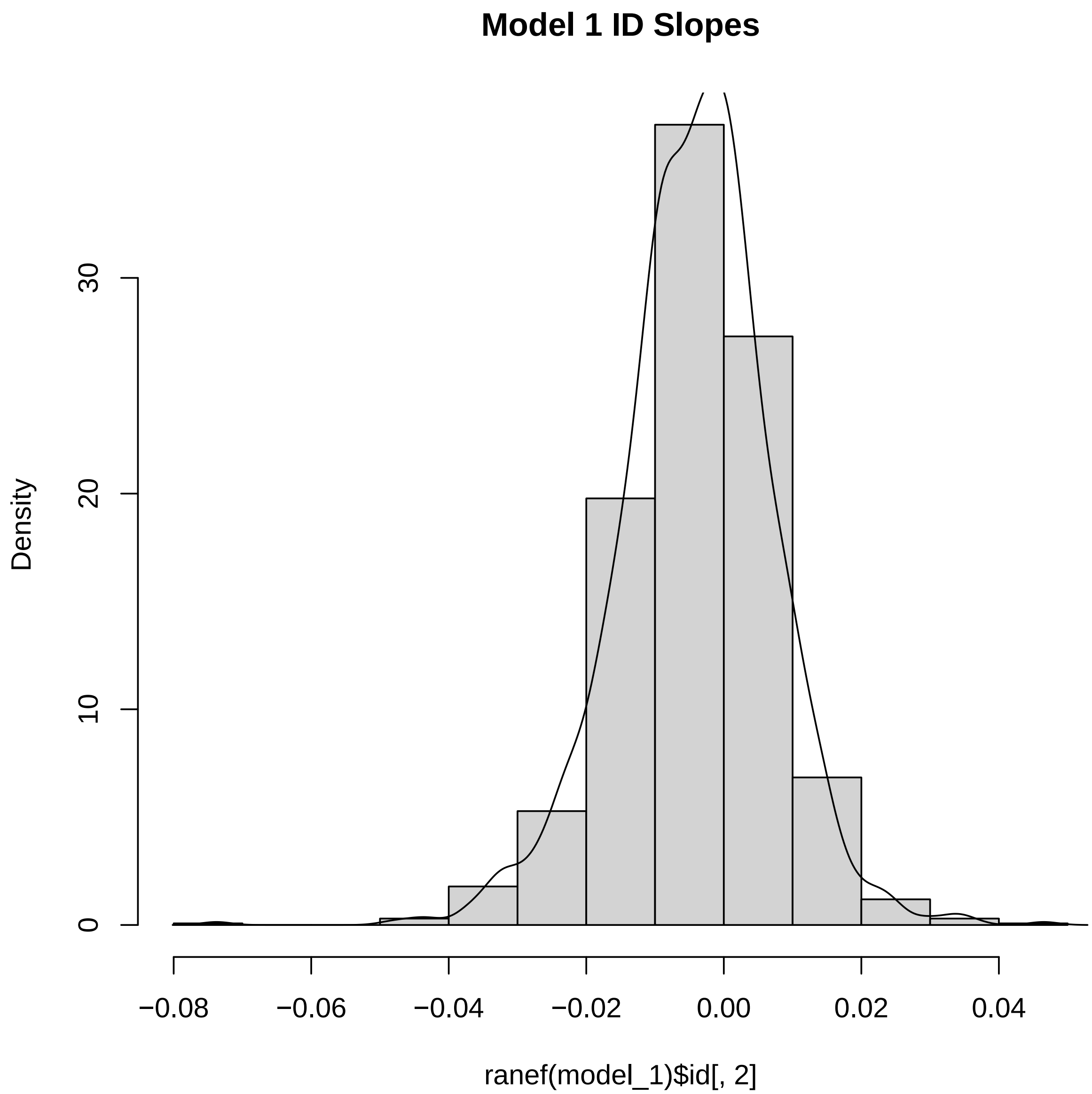
```
## Linear mixed model fit by REML ['lmerMod']
## Formula: fev ~ year * asthma + mht + sex + age_quartile + (1 | family/id) +
##     (age | id)
##    Data: .
##
## REML criterion at convergence: 1340.4
##
```

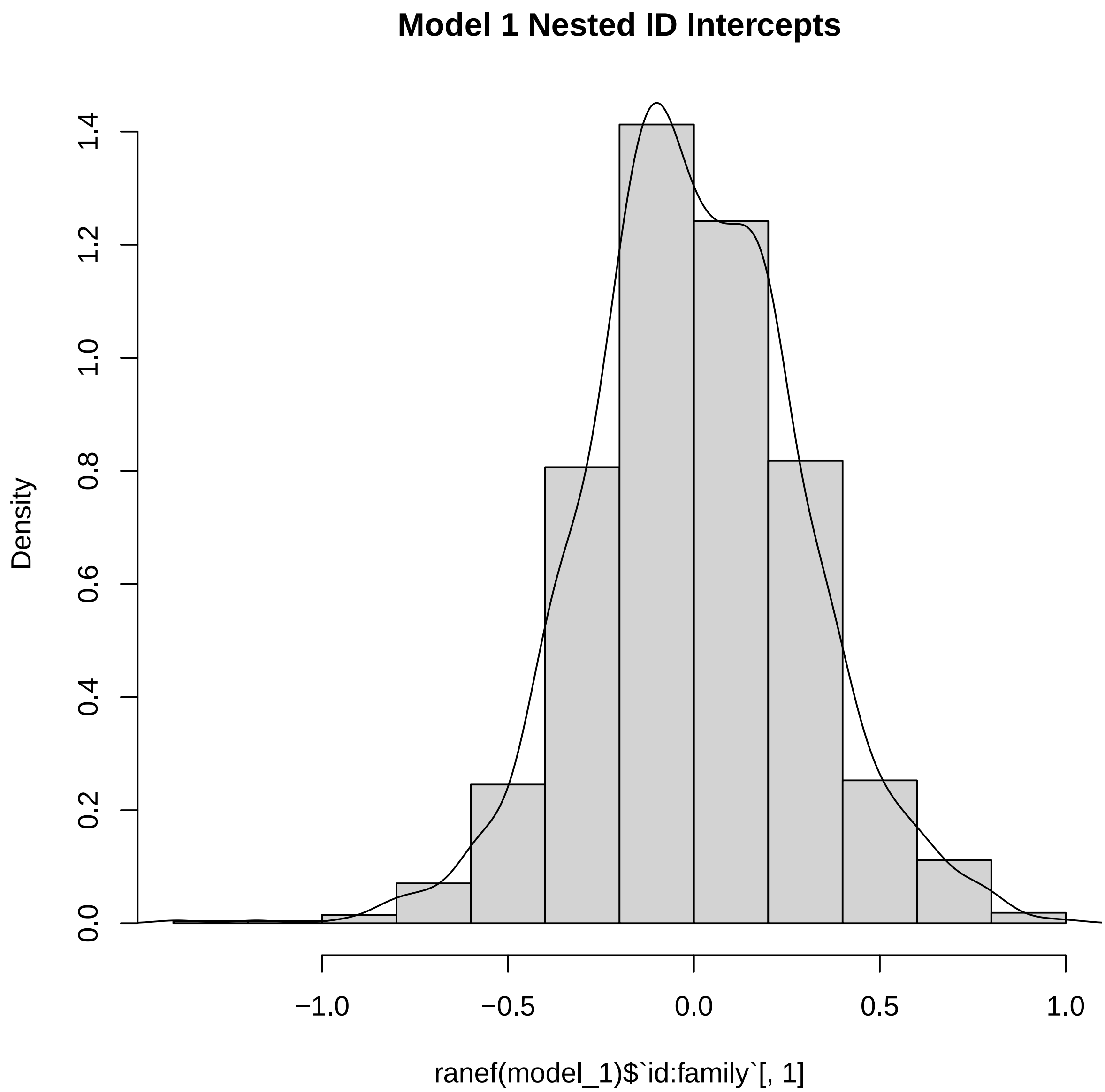
```
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -6.7620 -0.5136  0.0088  0.5123  4.2440
##
## Random effects:
##  Groups    Name        Variance  Std.Dev. Corr
##  id        (Intercept) 0.5281739 0.72676
##            age         0.0002641 0.01625  -1.00
##  id:family (Intercept) 0.1538233 0.39220
##  family    (Intercept) 0.0828996 0.28792
##  Residual              0.0321550 0.17932
## Number of obs: 7967, groups:  id, 1345; id:family, 1345; family, 1000
##
## Fixed effects:
##                          Estimate Std. Error t value
## (Intercept)             -4.945747   0.407534 -12.136
## year                    -0.018047   0.001118 -16.136
## asthma                  -0.204383   0.051734  -3.951
## mht                      0.129266   0.005875  22.003
## sexFemale               -0.342048   0.041707  -8.201
## age_quartile36.8-52.5   -0.163446   0.017697  -9.236
## age_quartile52.6-68.9   -0.353811   0.021471 -16.478
## age_quartile68.91-101.8 -0.464883   0.025299 -18.375
## year:asthma              0.007144   0.003659   1.953
##
## Correlation of Fixed Effects:
##             (Intr) year   asthma mht    sexFml a_36.8 a_52.6 a_68.9
## year         0.107
## asthma       0.025  0.053
## mht         -0.998 -0.112 -0.038
## sexFemale   -0.772 -0.056 -0.030  0.748
## a_36.8-52.5 -0.101 -0.049  0.024  0.068  0.040
## a_52.6-68.9 -0.167 -0.083  0.030  0.133  0.074  0.781
## a_68.91-101 -0.189 -0.145  0.027  0.158  0.092  0.659  0.834
## year:asthma -0.012 -0.295 -0.284  0.016  0.005 -0.044 -0.031 -0.011
## optimizer (nloptwrap) convergence code: 0 (OK)
## unable to evaluate scaled gradient
## Model failed to converge: degenerate  Hessian with 1 negative eigenvalues
```

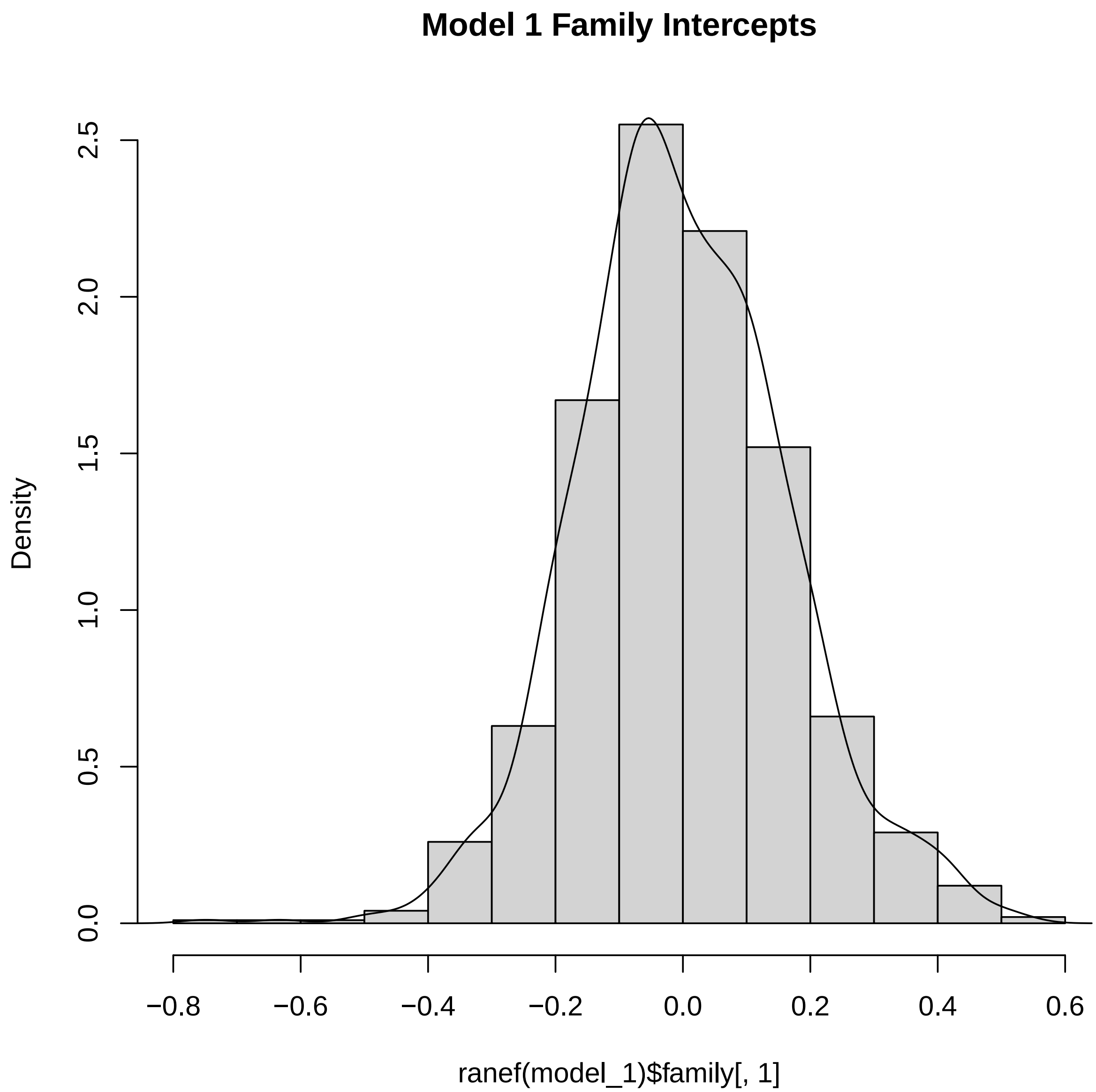

Model 1 Residuals
Error
0.5
0.0
−0.5
−1.0
0
2000
4000
6000
8000
Index


0.5
0.0
−0.5
−1.0
Sample
−4
−2
0
2
4
Theoretical


Model 1 ID Intercepts
Density
0.0
0.2
0.4
0.6
0.8
−2
−1
0
1
2
3
ranef(model_1)$id[, 1]


Model 1 ID Slopes
Density
0
10
20
30
−0.08
−0.06
−0.04
−0.02
0.00
0.02
0.04
ranef(model_1)$id[, 2]


Model 1 Nested ID Intercepts
Density
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
−1.0
−0.5
0.0
0.5
1.0
ranef(model_1)$`id:family`[, 1]



The model has random and normally distributed residuals, along with normally distributed random effects. It would be useful to consider the performance of a model based on uncorrelated slopes and intercepts, which was done below.

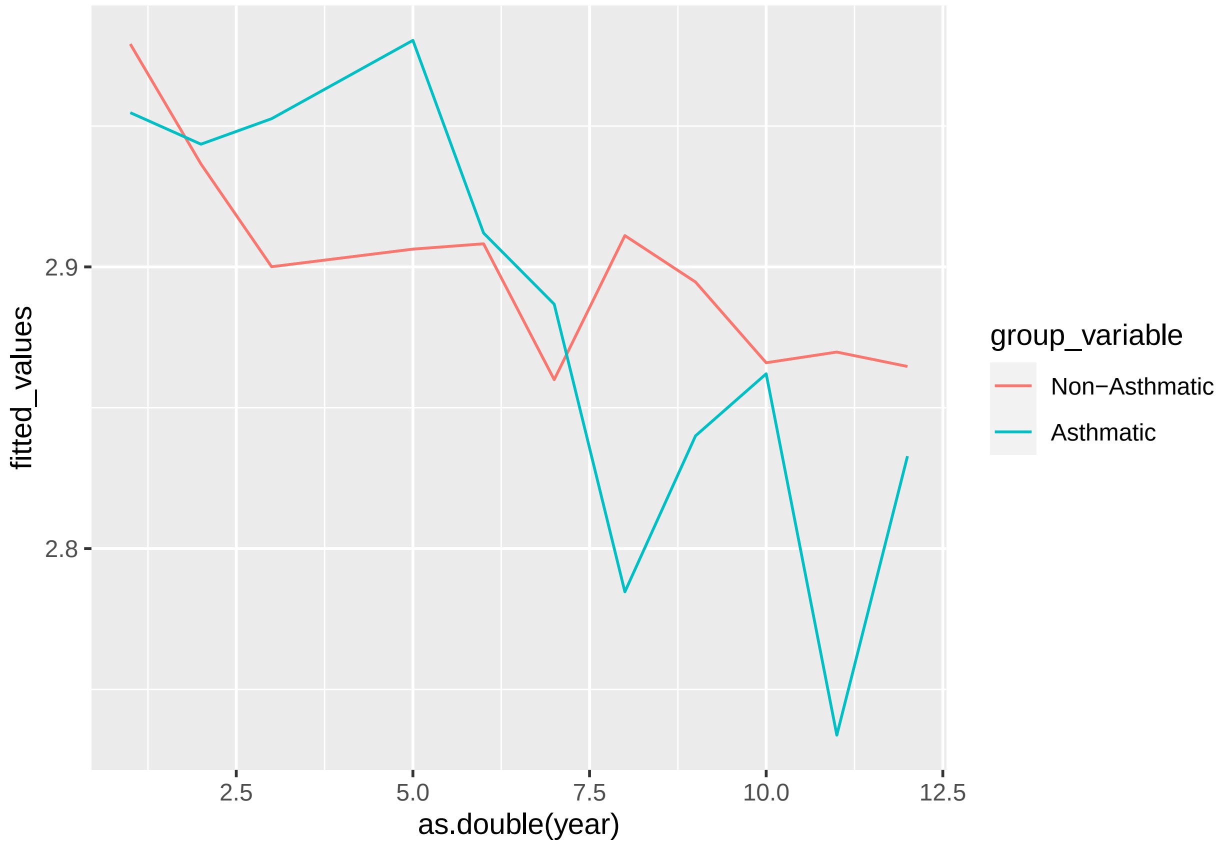
```
model_1b =
  family_asthma_long %>%
  lmer(fev ~ year * asthma + # Main Coefficients of Interest
       mht + sex  + age_quartile + # independent controls
       (1 | family / id) + (age || id) , data = .) # double bar = uncorrelated intercept/slope

anova(model_1,model_1b)
```

```
## refitting model(s) with ML (instead of REML)
```

```
## Data: .
## Models:
## model_1b: fev ~ year * asthma + mht + sex + age_quartile + (1 | family/id) + ((1 | id) + (0 + age | i
## model_1: fev ~ year * asthma + mht + sex + age_quartile + (1 | family/id) + (age | id)
##          npar    AIC    BIC  logLik deviance  Chisq Df Pr(>Chisq)
## model_1b   14 1659.0 1756.8 -815.51   1631.0
## model_1    15 1304.9 1409.7 -637.47   1274.9 356.08  1  < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The model including the covariance term performs significantly better than the model, with a lower AIC and extremely significant chi-square test statistic. In other words, the slopes and intercepts of the subject-by-age coefficients are not independent. Thus, the term ought to be included in modeling this data set, as doing so will provide more accurate and useful predictions. It makes intuitive sense to include the covariance structure, because as discovered before many of the variables in the data are highly correlated with one another.

```r
df1 <- family_asthma_long %>%
  filter(!is.na(sex), !is.na(asthma), !is.na(fev), !is.na(age_quartile),
         !is.na(mht)) %>%
  mutate(asthma = factor(asthma, labels = c("0" = "Non-Asthmatic", "1" = "Asthmatic")))

# Create a new dataframe for the fitted values
df_fitted1 <- data.frame(year = as.integer(df1$year),
                         fitted_values = predict(model_1),
                         group_variable = df1$asthma %>% factor(labels =
                             c("0" = "Non-Asthmatic", "1" = "Asthmatic")))


p <- family_asthma_long %>%
  mutate(group = interaction(sex, asthma) %>% factor(labels =
    c("Male.0" = "Men", "Male.1" = "Asthmatic Men", "Female.0" = "Women",
    "Female.1" = "Asthmatic Women"))) %>%
  filter(!is.na(group)) %>%
  ggplot(aes(year, fev, color = group)) +
  geom_point()


# Add lines to the scatterplot
df_fitted1 %>%
  group_by(group_variable, year) %>%
  reframe(fitted_values = mean(fitted_values, na.rm = TRUE)) %>%
  ggplot(
    aes(x = as.double(year), y = fitted_values, color = group_variable)
  ) +
geom_line()
```

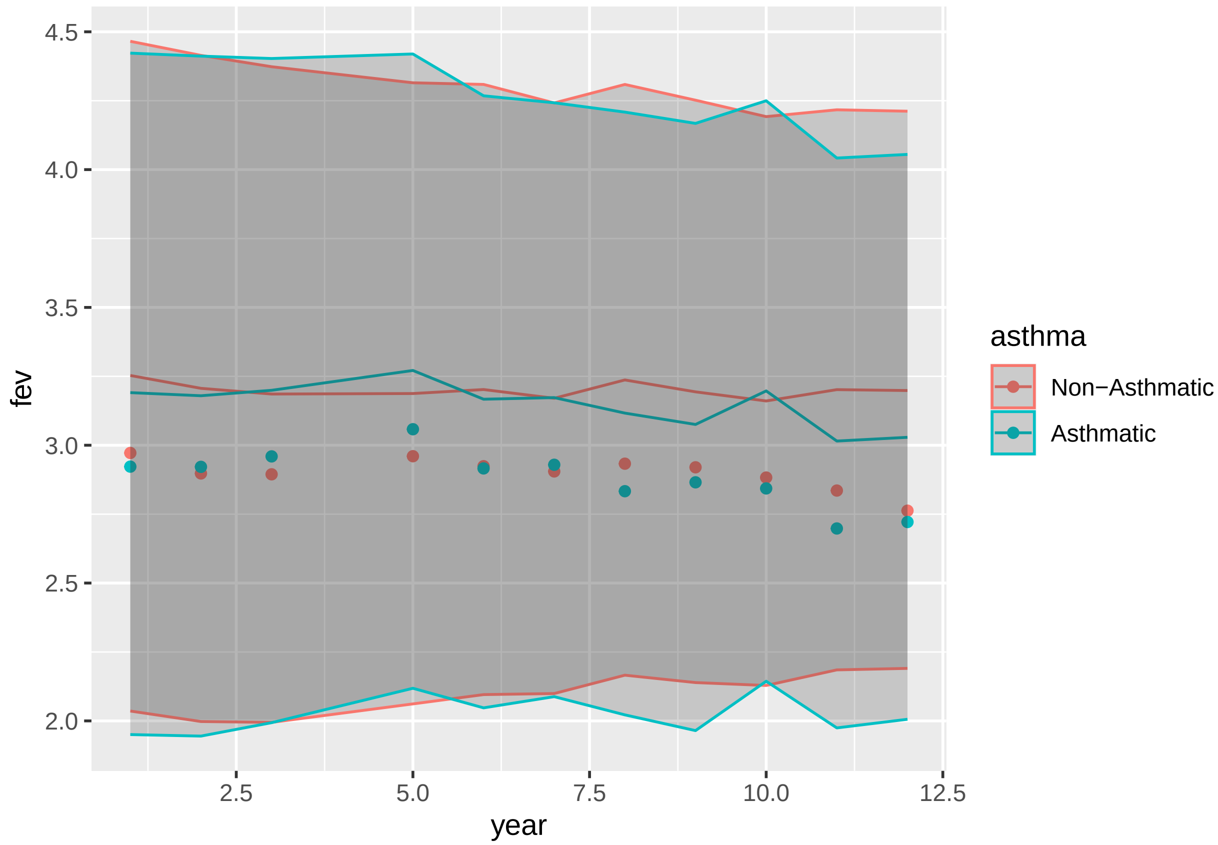



```r
# Use predictInterval to get the fitted values and confidence intervals
df_fitted1 <- predictInterval(model_1, newdata = df1, level = 0.95)
```

```
## Warning in predictInterval(model_1, newdata = df1, level = 0.95): newdata is tbl_df or tbl object fr
##               coerced to a data.frame

## Warning: executing %dopar% sequentially: no parallel backend registered
```

```r
# Add the fitted values and confidence intervals to the original dataframe
df1$Fitted <- df_fitted1$fit
df1$LowerCI <- df_fitted1$lwr
df1$UpperCI <- df_fitted1$upr

# Create the scatterplot with geom_point
p <-
  df1 %>%
  group_by(asthma, year) %>%
  reframe(Fitted = mean(Fitted),
          fev = mean(fev),
          UpperCI = mean(UpperCI),
          LowerCI = mean(LowerCI)) %>%
  ggplot(aes(x = year, y = fev, group = asthma, color = asthma)) +
  geom_point()
# Add the fitted lines with geom_line
p <- p + geom_line(aes(y = Fitted))
```

```r
# Add the confidence intervals with geom_ribbon
p <- p + geom_ribbon(aes(ymin = LowerCI, ymax = UpperCI), alpha = 0.2)
p
```



2. Do asthmatic smokers have steeper rates of decline or lower levels of lung function than non-asthmatic smokers?

a) using the fixed effect variable ever smoke (smoke_ever)

```r
model_2 <-
  family_asthma_long %>%
  lmer(fev ~ year * asthma  * smoke_ever + # Main Coefficients of Interest
       mht + sex  + age_quartile + # independent controls
       (1 | family / id) + (age | id) , data = .) # single bar = correlated intercept/slope
tabled_model_2 = regression_table(model_2)
```

This model uses the categorical variable for whether or not the participant ever smoked. Said variable is interacted with the time and asthma so that an estimate for slope can be calculated. Interacting all three of these terms would allow for interpretation on: i) the rate of change of FEV over time for asthmatics, ii) the rate of change of FEV over time for smokers, iii) the rate of change of FEV over time for asthmatic smokers, The control variables remain as they are known to have a significant effect on FEV and should thus be accounted for.

b) using the time dependent variable for smoking (smk)

```r
model_3 =
  family_asthma_long %>%
  lmer(fev ~ year * asthma * smk + # Main Coefficients of Interest
       mht + sex + age_quartile + # independent controls
       (1 | family / id) + (age | id) , data = .) # single bar = correlated intercept/slope
tabled_model_3 = regression_table(model_3)
```

This model interacts time and asthma with the time-dependent variable smk, which indicates whether or not the subject smoked in the current year. The resulting coefficients are similar to that of a previous model, but may lead to different interpretations as "smk" can change for individuals. c) using the current number of cigarettes per day (cncig)

```r
model_4 =
  family_asthma_long %>%
  lmer(fev ~ year * asthma*cncig_quartile + # Main Coefficients of Interest
       mht + sex  + age_quartile + # independent controls
       (1 | family / id) + (age | id) , data = .) # single bar = correlated intercept/slope
tabled_model_4 = regression_table(model_4)
```

The final approach factored in the number of cigarettes smoked by the participants per day. With this model we will get several groups of categorical variables as cncig_quartile is interacting with both year and time. Analysis will be done on i) significance of smoking frequency, ii) the rate of change of FEV with respect to smoking frequency, iii) the change in FEV levels for asthmatics according to smoking frequency (note this is not over time), and iv) the rate of change of FEV for asthmatics with respect to smoking frequency.

Question 2 Analysis

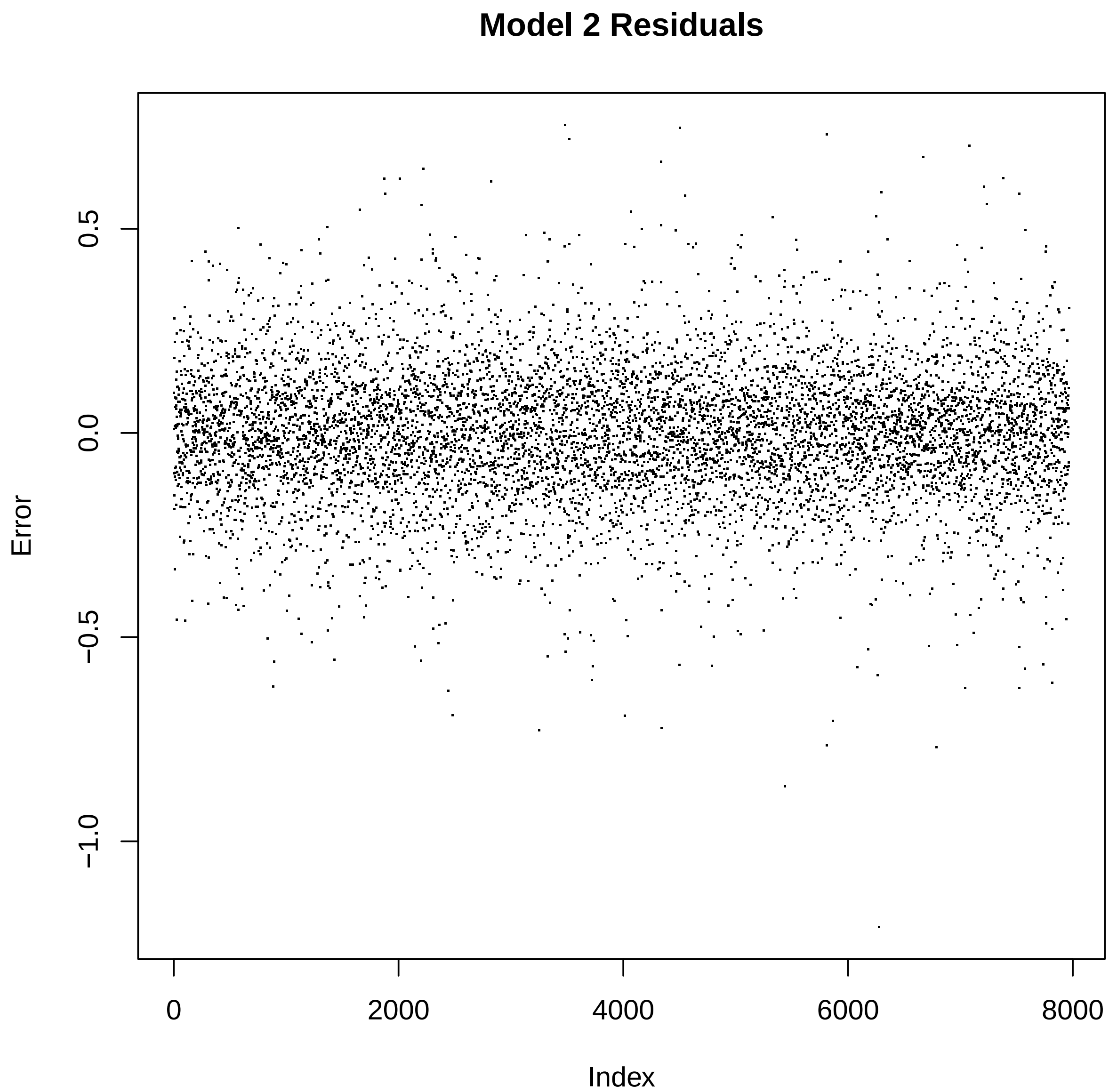
Summary of Results

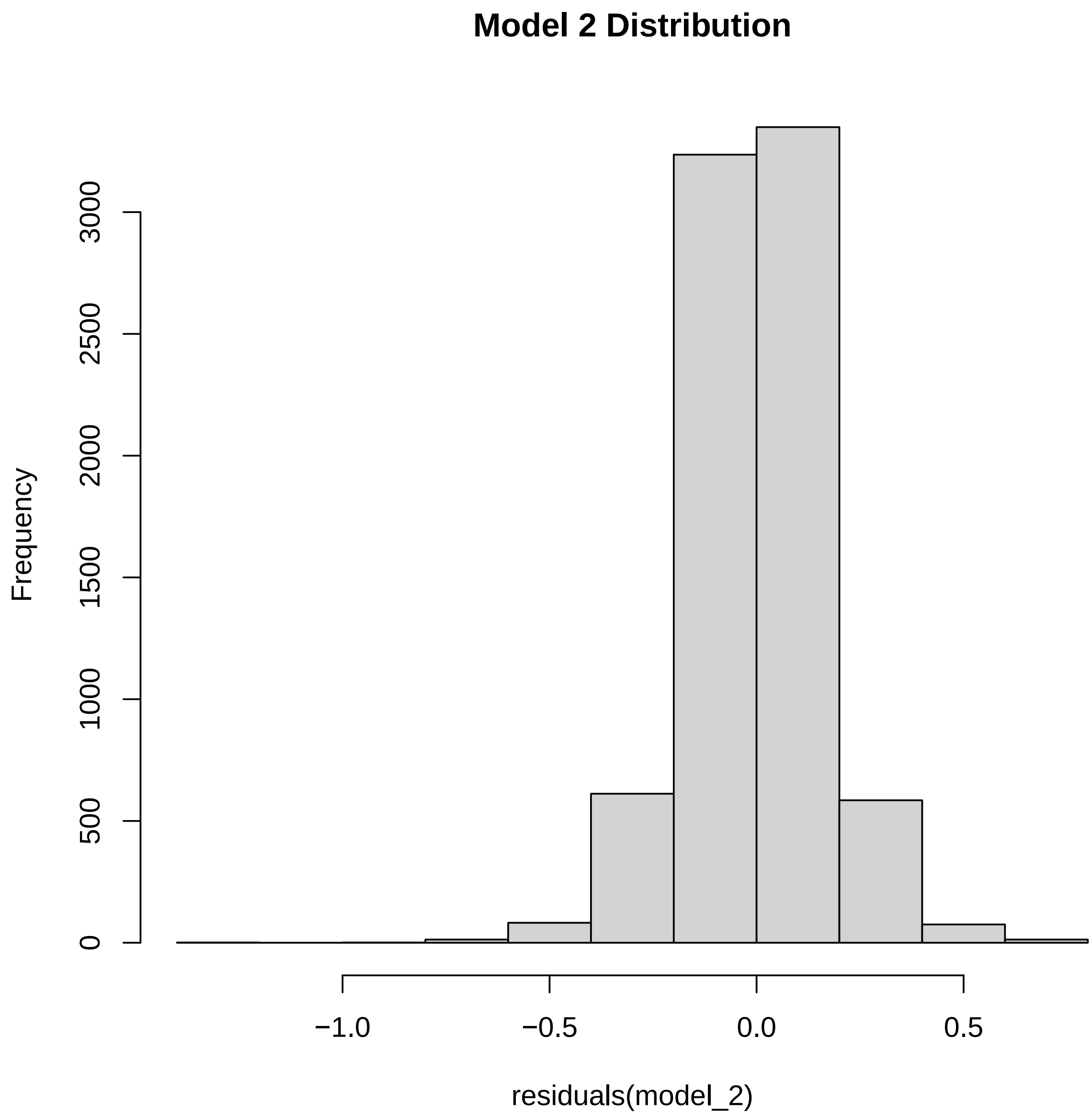
Characteristic	**Beta (SE)**	**Beta (SE)**	**Beta (SE)**
Time (years)	-0.017*** (0.001)	-0.018*** (0.001)	-0.017*** (0.001)
Asthmatic	-0.268*** (0.064)	-0.218*** (0.054)	-0.215*** (0.054)
Ever Smoked	-0.064 (0.034)		
Average Height	0.129*** (0.006)	0.129*** (0.006)	0.131*** (0.006)
sex	NA*** (NA)	NA*** (NA)	NA*** (NA)
Male	—	—	—
Female	-0.347	-0.345	-0.332
Age Quartile	NA*** (NA)	NA*** (NA)	NA*** (NA)
20.0-36.7	—	—	—
36.8-52.5	-0.164	-0.165	-0.181
52.6-68.9	-0.356	-0.357	-0.379
68.91-101.8	-0.469	-0.469	-0.498
Time (years) * Asthmatic	0.009* (0.005)	0.009* (0.004)	0.010* (0.004)
Time (years) * Ever Smoked	-0.003 (0.002)		
Asthmatic * Ever Smoked	0.177 (0.107)		
Time (years) * Asthmatic * Ever Smoked	-0.006 (0.008)		
Smoke this year		-0.027 (0.016)	
Time (years) * Smoke this year		0.001 (0.002)	
Asthmatic * Smoke this year		0.048 (0.053)	
Time (years) * Asthmatic * Smoke this year		-0.005 (0.008)	
Smoker Frequency Quantile			NA (NA)

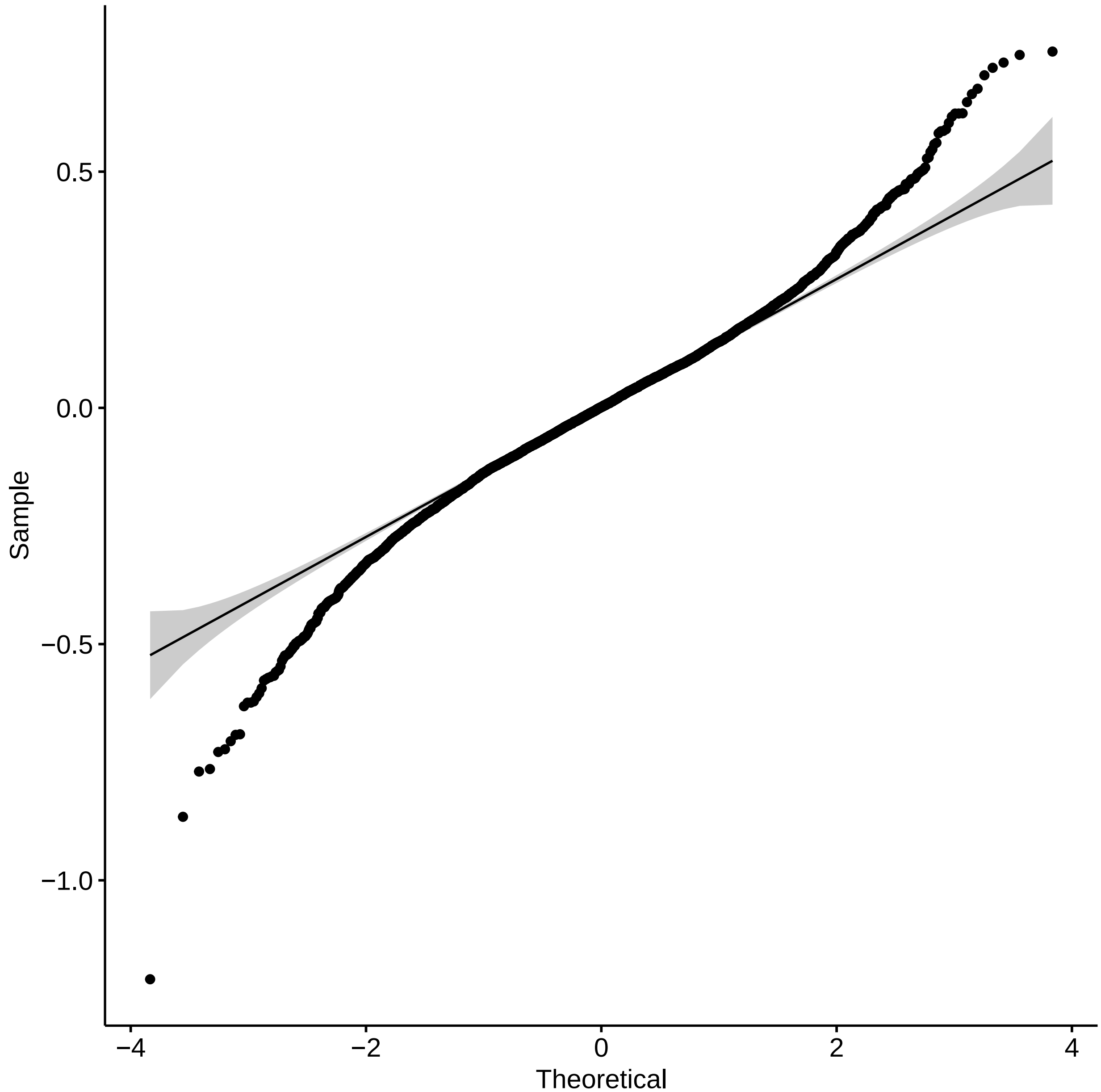
Characteristic	Beta (SE)	Beta (SE)	Beta (SE)
Lowest 25%			—
25%-50%			-0.004
50%-75%			-0.007
Highest 25%			0.033
Time (years) * Smoker Frequency Quantile			NA (NA)
Time (years) * 25%-50%			-0.005
Time (years) * 50%-75%			-0.007
Time (years) * Highest 25%			0.000
Asthmatic * Smoker Frequency Quantile			NA (NA)
Asthmatic * 25%-50%			0.036
Asthmatic * 50%-75%			0.004
Asthmatic * Highest 25%			-0.058
Time (years) * Asthmatic * Smoker Frequency Quantile			NA (NA)
Time (years) * Asthmatic * 25%-50%			-0.023
Time (years) * Asthmatic * 50%-75%			-0.003
Time (years) * Asthmatic * Highest 25%			-0.012

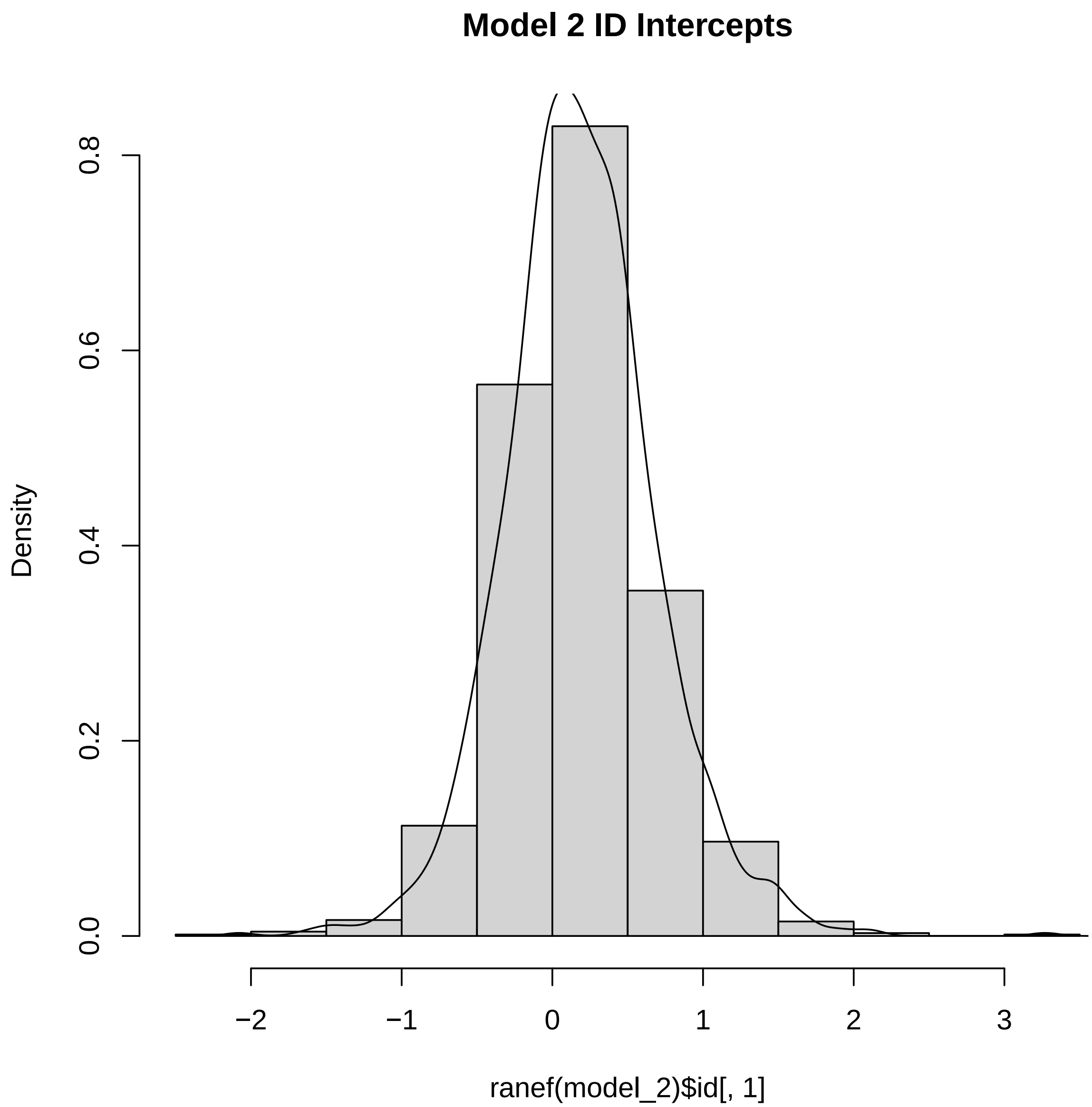
Overall we see the results are pretty similar between the three models. All of them find increased time, being asthmatic, or being female as significant factors in lowering FEV levels. They all show that taller individuals have significantly higher levels of lung function than shorter individuals, and that age is a significant factor in reducing lung function, an effect that becomes stronger at successively older ages. All models found the interaction between time and asthma to be significant (despite relatively large respective standard errors), albeit in the positive direction which is counter-intuitive. This suggests that being asthmatic leads to increases lung function over time. Model 2, based on the categorical variable representing whether or not the patient has ever smoked, found said variable to be negative. This means that people who have smoked at any point have lower FEV than those who have not, but this result is not statistically significant. Interestingly, the positive estimate for the interaction between asthma and smoking suggests that asthmatic smokers have higher levels of FEV than non-asthmatic smokers, but this result is also insignificant with a very high standard error. *The slope estimate for asthmatic smokers was -0.006, with another large standard error, making it insignificant despite its implication of reduced lung function over time for asthmatic smokers.* Model 3, which uses time-dependent variable “smoke this year” to indicate whether the participant smoked at the given time point, found that whether or not the individual smoked at all during the year was insignificant in reducing FEV levels. Additionally, all the interaction terms (i.e. smoke versus time, asthmatic smokers, and asthmatic smokers over time) were insignficant, *suggesting that smoking in the current year has no statistically significant effect on FEV levels over time for asthmatics or non-asthmatics.* Model 4 uses the categorical variable for the number of cigarettes smoked by the participant. Neither the variable itself nor any of its interactions with time or asthma were found to be significant, which implies that smoking frequency does not significantly impact FEV levels over time either, regardless of asthma status.
As a general answer to the research question, the models suggest that being asthmatic leads to lower levels of lung function, but whether or not the participant smoked is insignificant in affecting FEV both over time or for initial levels. That is, there is insufficient statistical evidence to conclude that asthmatic smokers have lower levels or steeper rates of decline in lung function than non-asthmatic smokers.

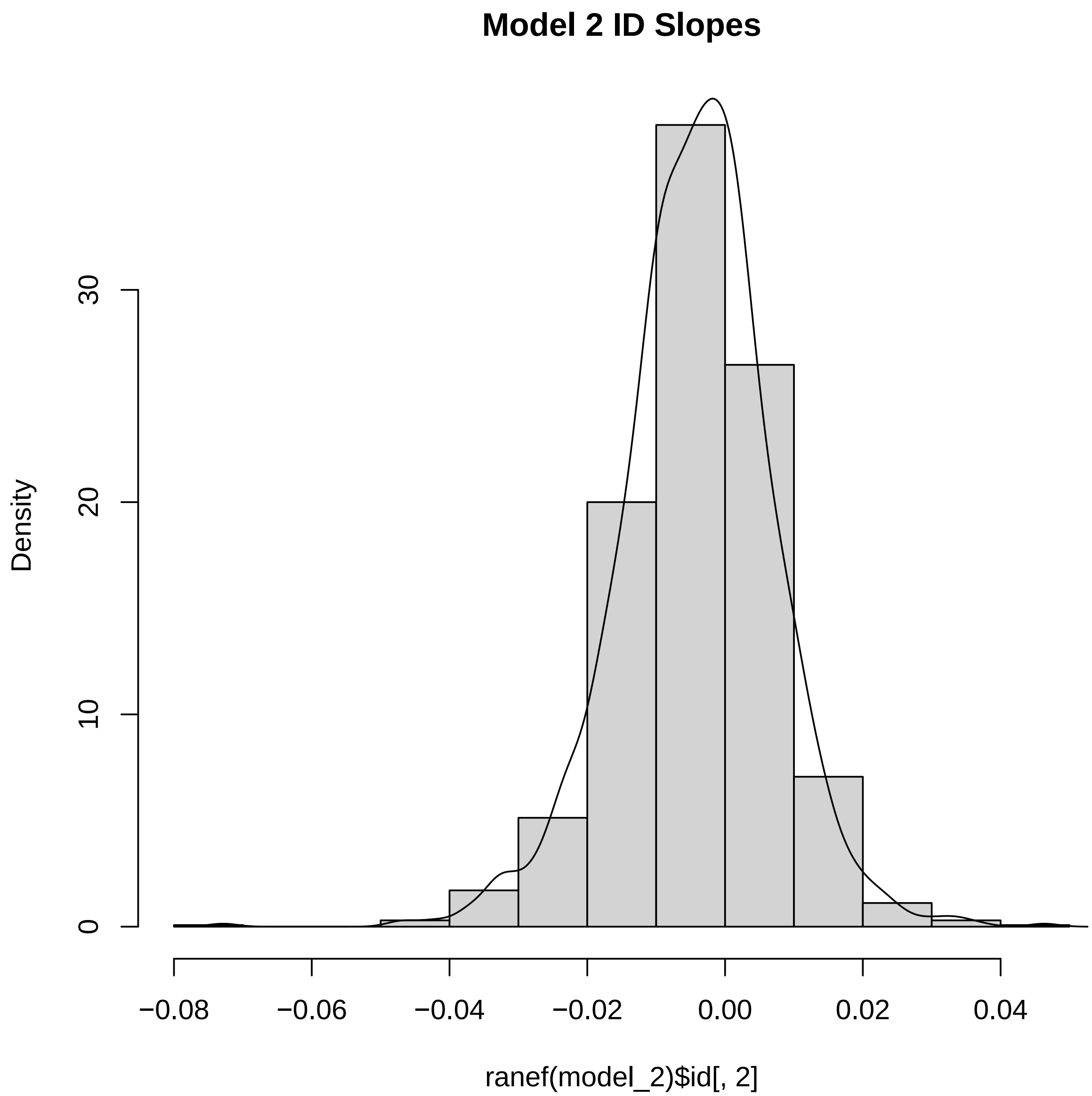
Model 2 Diagnostics

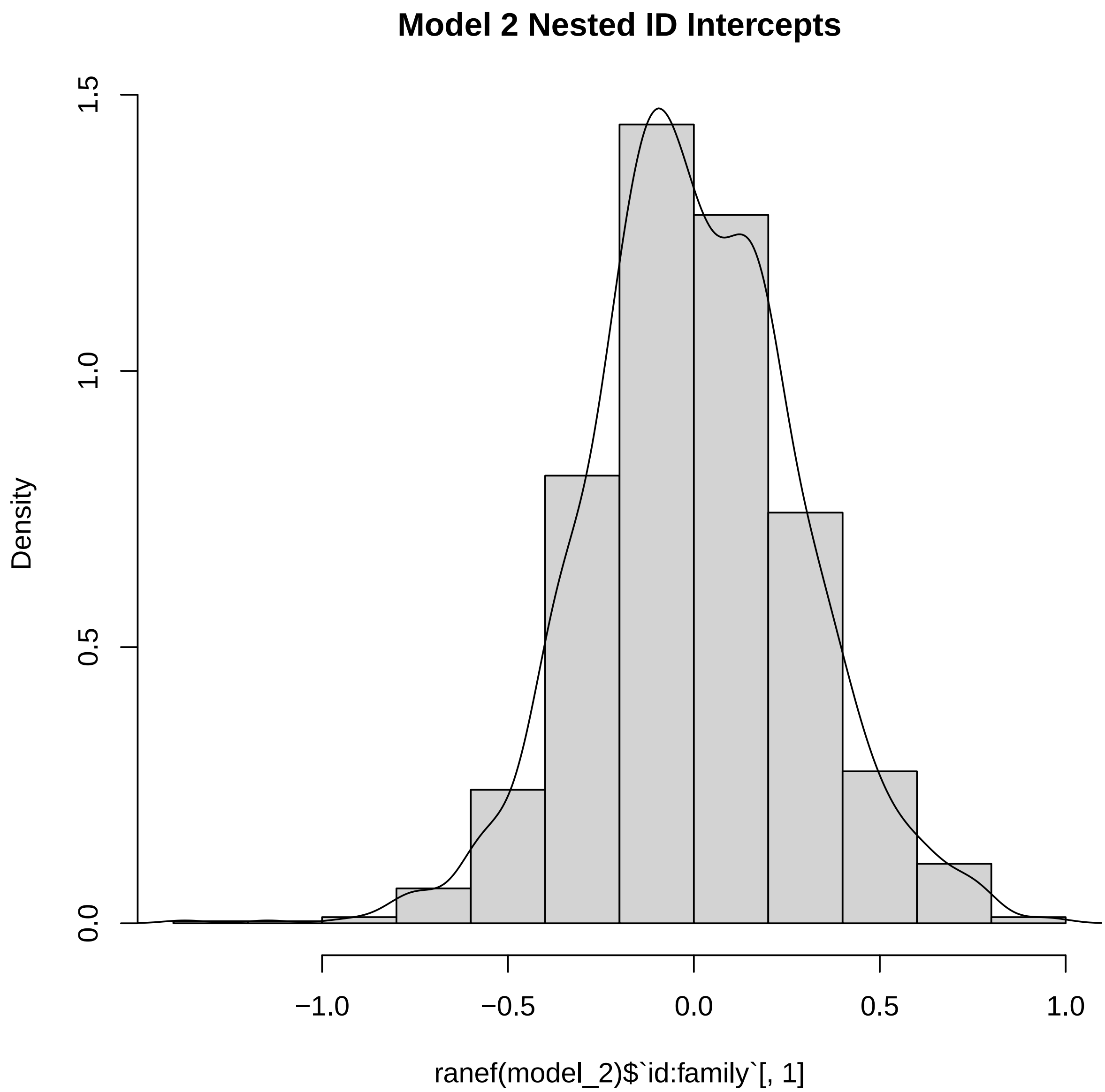

Model 2 Residuals
Error
0.5
0.0
−0.5
−1.0
0
2000
4000
6000
8000
Index

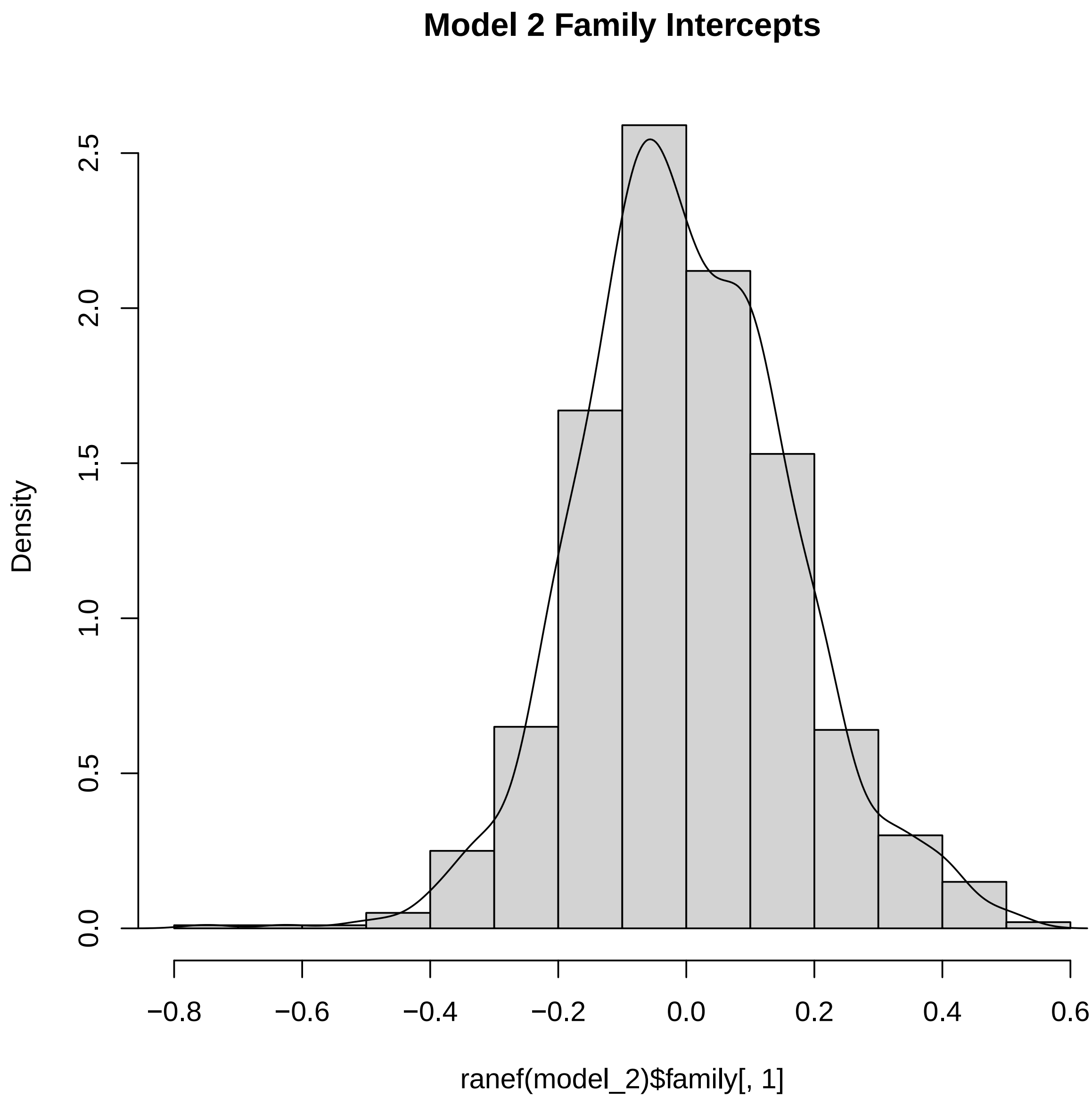

Model 2 Distribution
Frequency
0
500
1000
1500
2000
2500
3000
−1.0
−0.5
0.0
0.5
residuals(model_2)


0.5
0.0
−0.5
−1.0
Sample
−4
−2
0
2
4
Theoretical

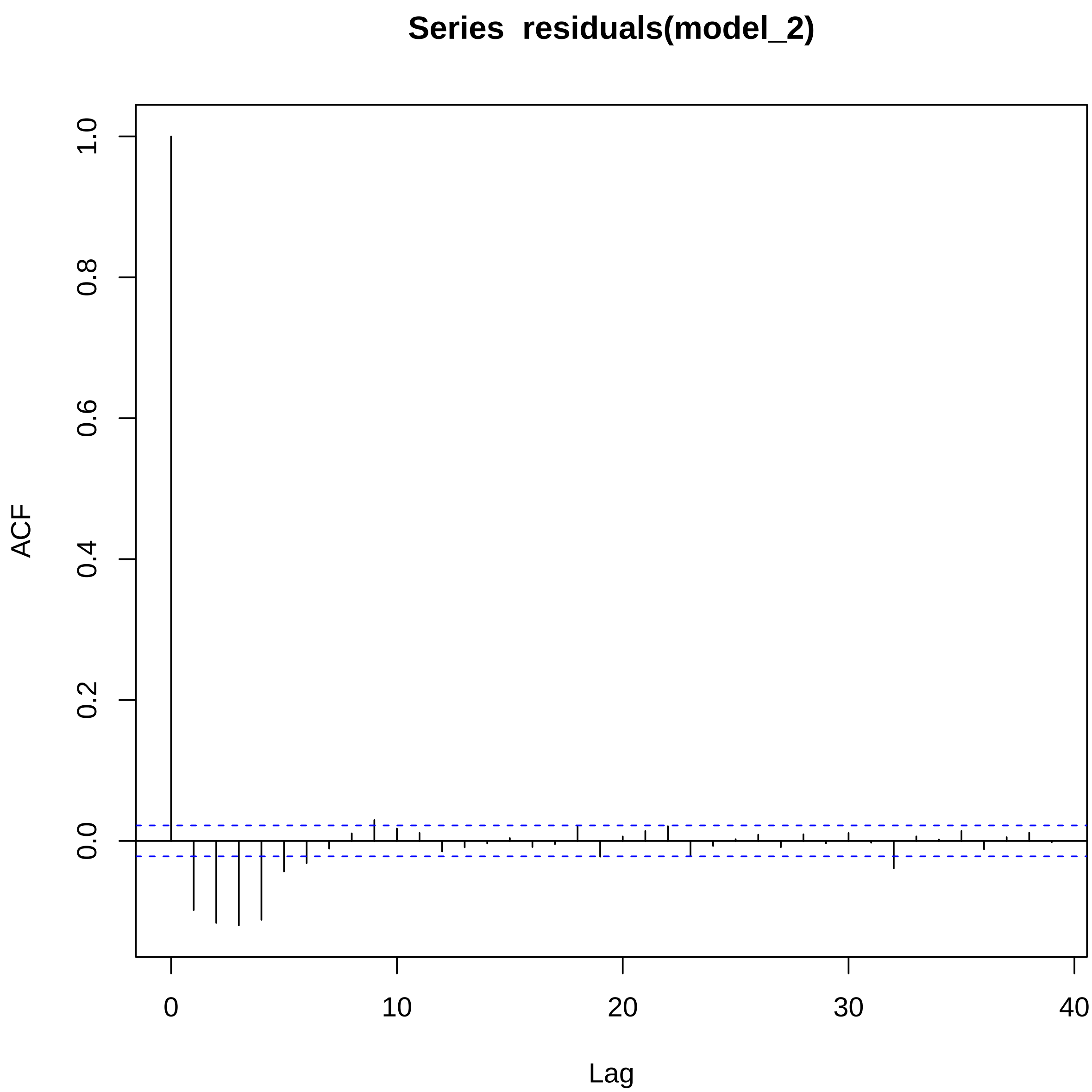

Model 2 ID Intercepts
Density
0.0
0.2
0.4
0.6
0.8
−2
−1
0
1
2
3
ranef(model_2)$id[, 1]


Model 2 ID Slopes
Density
0
10
20
30
−0.08
−0.06
−0.04
−0.02
0.00
0.02
0.04
ranef(model_2)$id[, 2]


Model 2 Nested ID Intercepts
Density
0.0
0.5
1.0
1.5
−1.0
−0.5
0.0
0.5
1.0
ranef(model_2)$`id:family`[, 1]

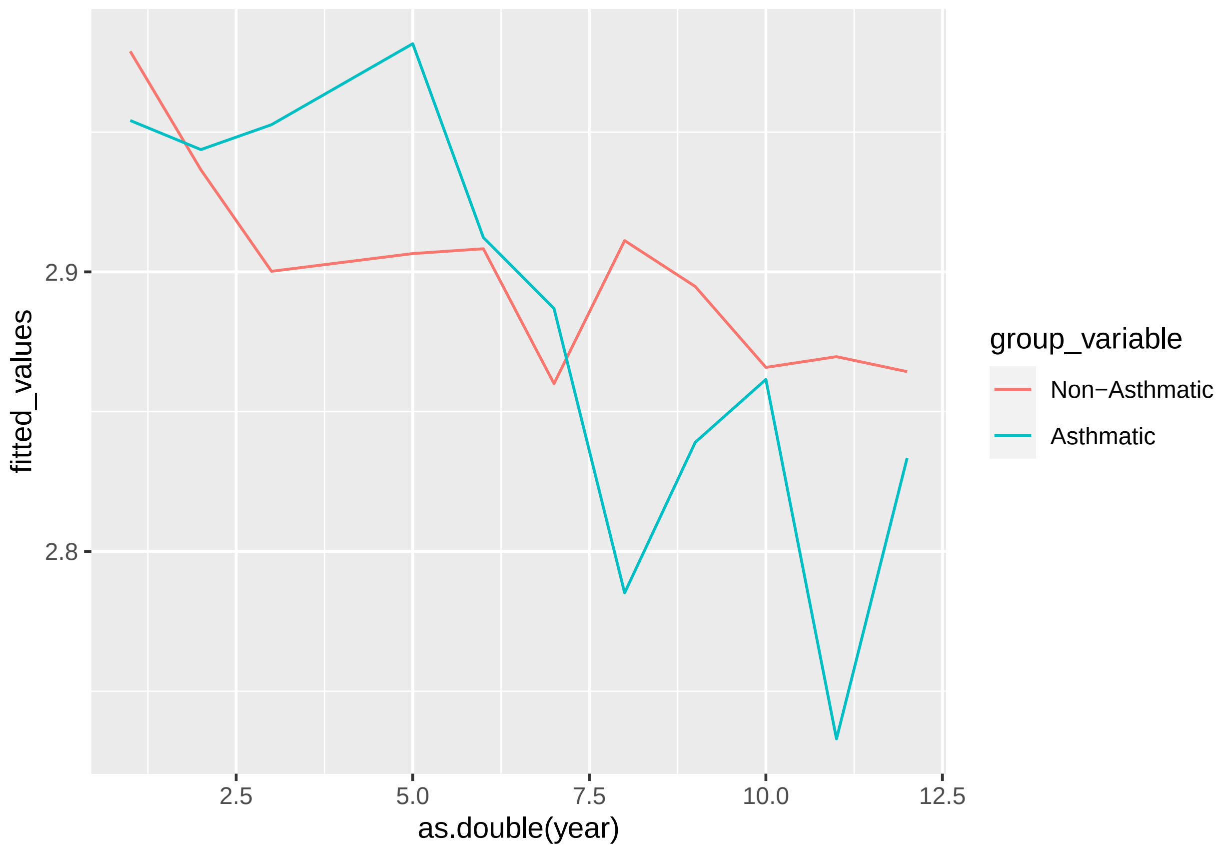



```
##
##  Box-Pierce test
##
## data:  residuals(model_2)
## X-squared = 76.654, df = 1, p-value < 2.2e-16
```




```
df2 <- family_asthma_long %>%
  filter(!is.na(sex), !is.na(asthma), !is.na(fev), !is.na(age_quartile),
         !is.na(mht)) %>%
  mutate(asthma = factor(asthma, labels = c("0" = "Non-Asthmatic", "1" = "Asthmatic")))

# Create a new dataframe for the fitted values
df_fitted2 <- data.frame(year = as.integer(df2$year),
                         fitted_values = predict(model_2),
                         group_variable = df2$asthma %>% factor(labels =
                             c("0" = "Non-Asthmatic", "1" = "Asthmatic")))


p <- family_asthma_long %>%
```

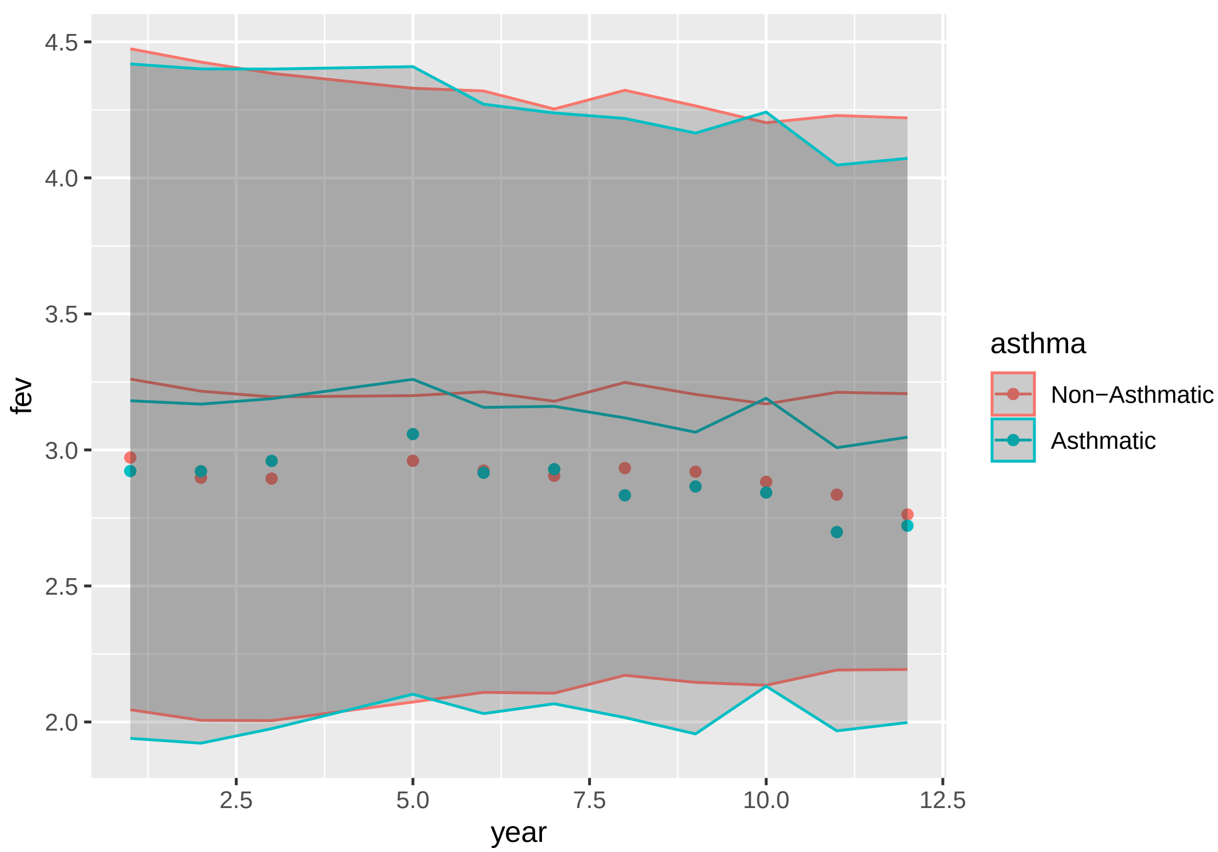
```r
  mutate(group = interaction(sex, asthma) %>% factor(labels =
    c("Male.0" = "Men", "Male.1" = "Asthmatic Men", "Female.0" = "Women",
    "Female.1" = "Asthmatic Women"))) %>%
  filter(!is.na(group)) %>%
  ggplot(aes(year, fev, color = group)) +
  geom_point()


# Add lines to the scatterplot
df_fitted2 %>%
  group_by(group_variable, year) %>%
  reframe(fitted_values = mean(fitted_values, na.rm = TRUE)) %>%
  ggplot(
    aes(x = as.double(year), y = fitted_values, color = group_variable)
  ) +
geom_line()
```



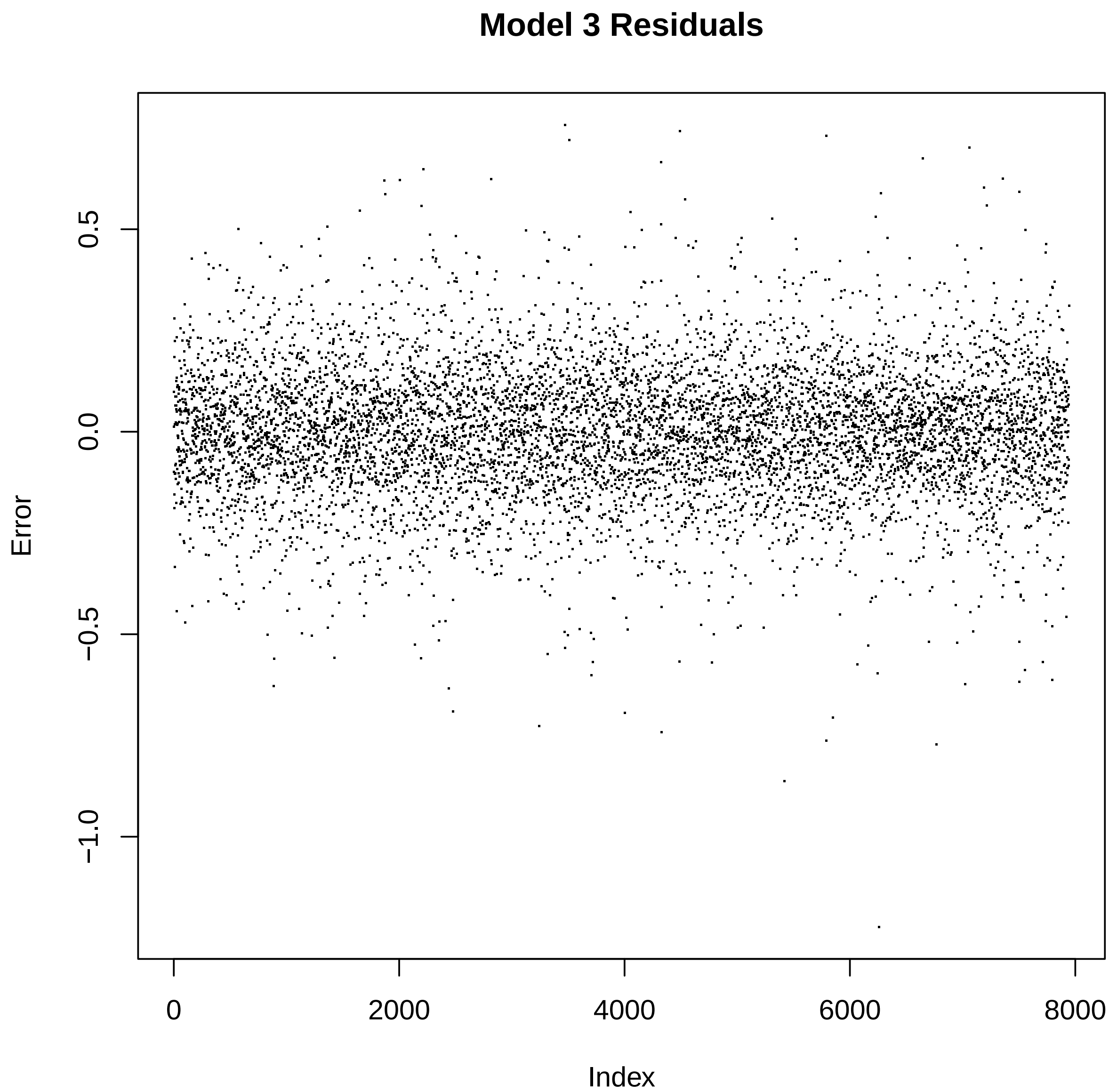

```r
# Use predictInterval to get the fitted values and confidence intervals
df_fitted2 <- predictInterval(model_2, newdata = df2, level = 0.95)
```

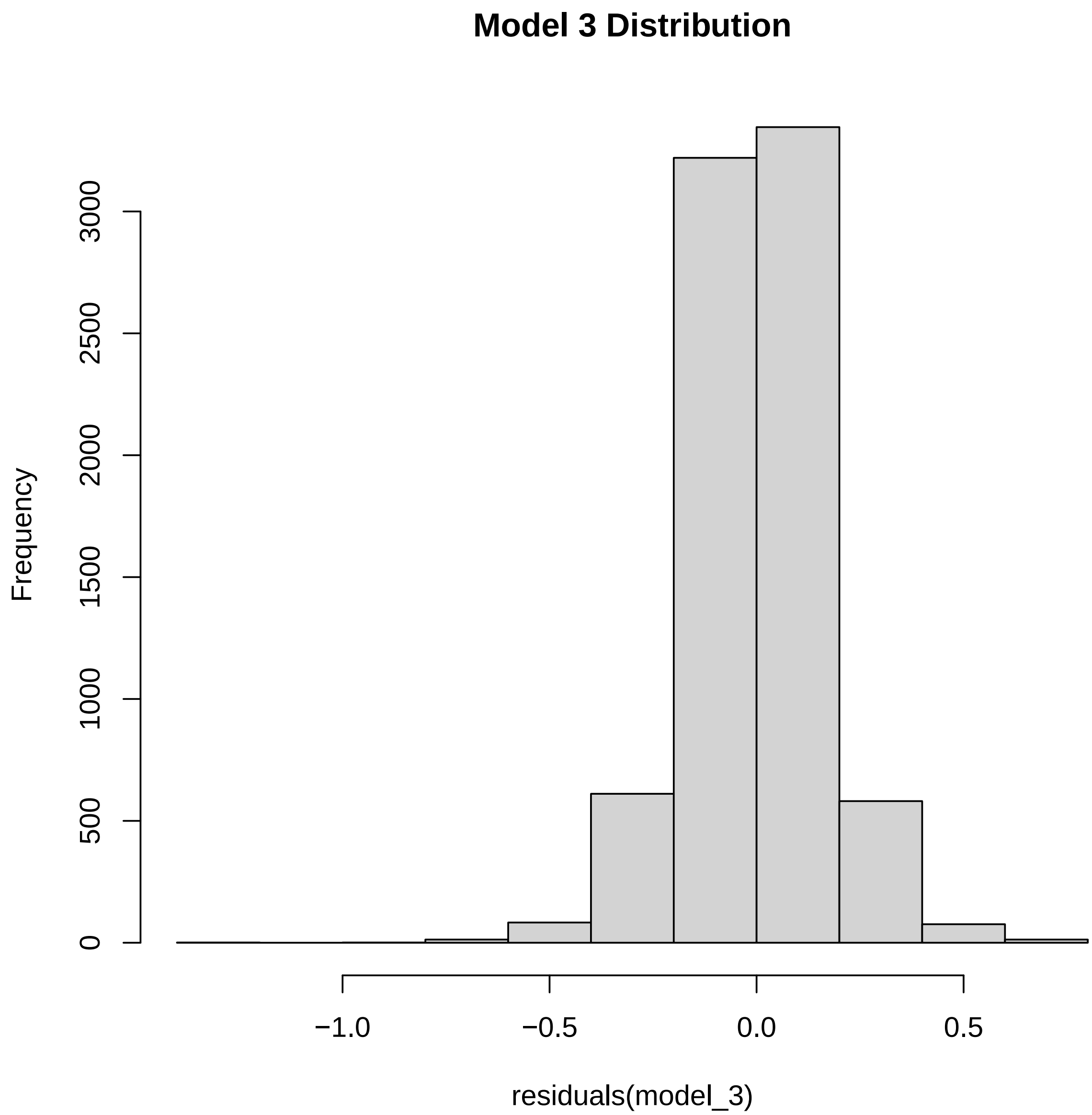
```
## Warning in predictInterval(model_2, newdata = df2, level = 0.95): newdata is tbl_df or tbl object fr
##                 coerced to a data.frame
```

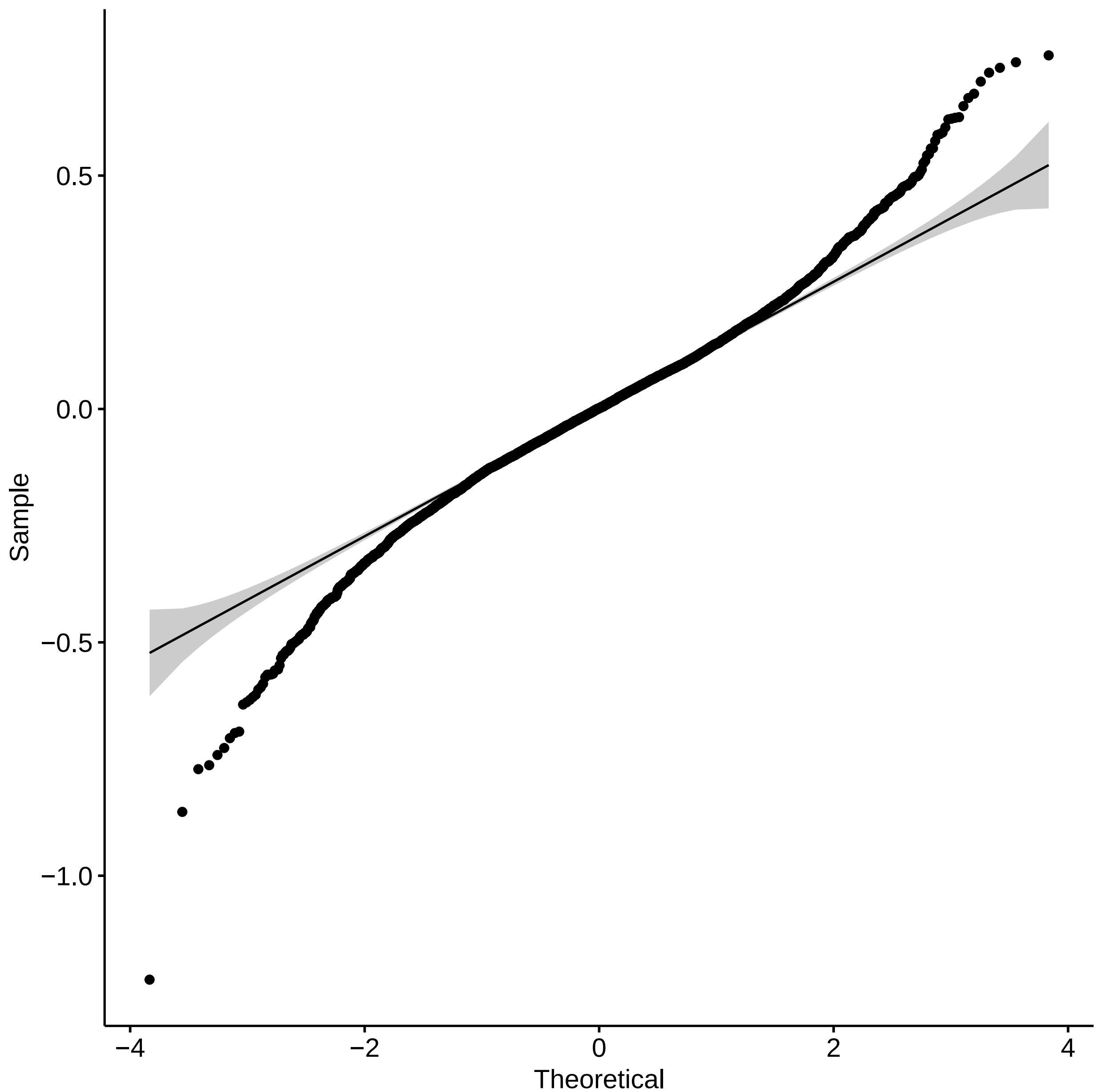
```r
# Add the fitted values and confidence intervals to the original dataframe
df2$Fitted <- df_fitted2$fit
df2$LowerCI <- df_fitted2$lwr
df2$UpperCI <- df_fitted2$upr

# Create the scatterplot with geom_point
p2 <-
  df2 %>%
  group_by(asthma, year) %>%
  reframe(Fitted = mean(Fitted),
          fev = mean(fev),
          UpperCI = mean(UpperCI),
          LowerCI = mean(LowerCI)) %>%
  ggplot(aes(x = year, y = fev, group = asthma, color = asthma)) +
  geom_point()
# Add the fitted lines with geom_line
p2 <- p2 + geom_line(aes(y = Fitted))
# Add the confidence intervals with geom_ribbon
p2 <- p2 + geom_ribbon(aes(ymin = LowerCI, ymax = UpperCI), alpha = 0.2)
p2
```



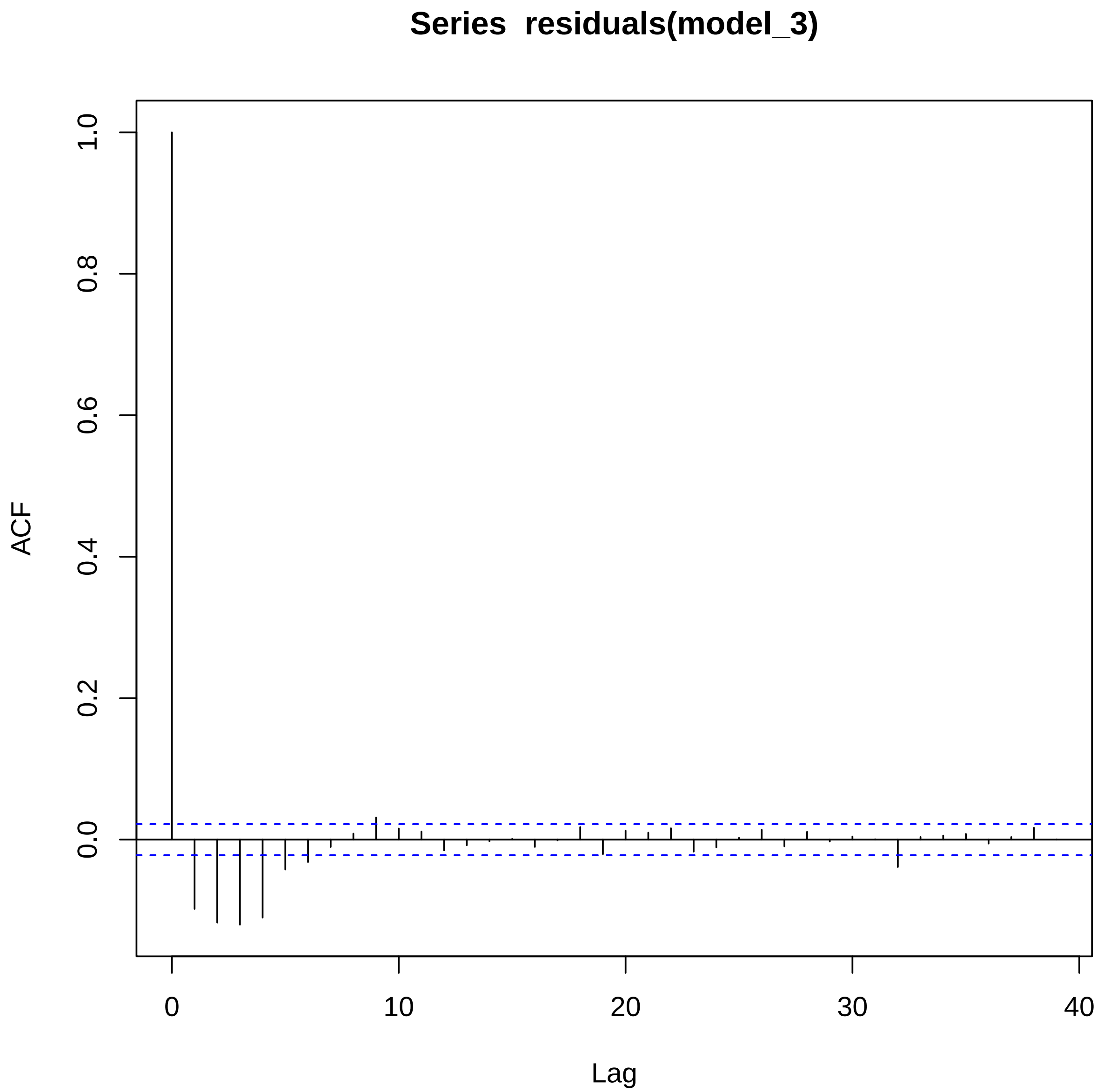
Model 3 Diagnostics


Model 3 Residuals
0.5
0.0
−0.5
−1.0
Error
0
2000
4000
6000
8000
Index

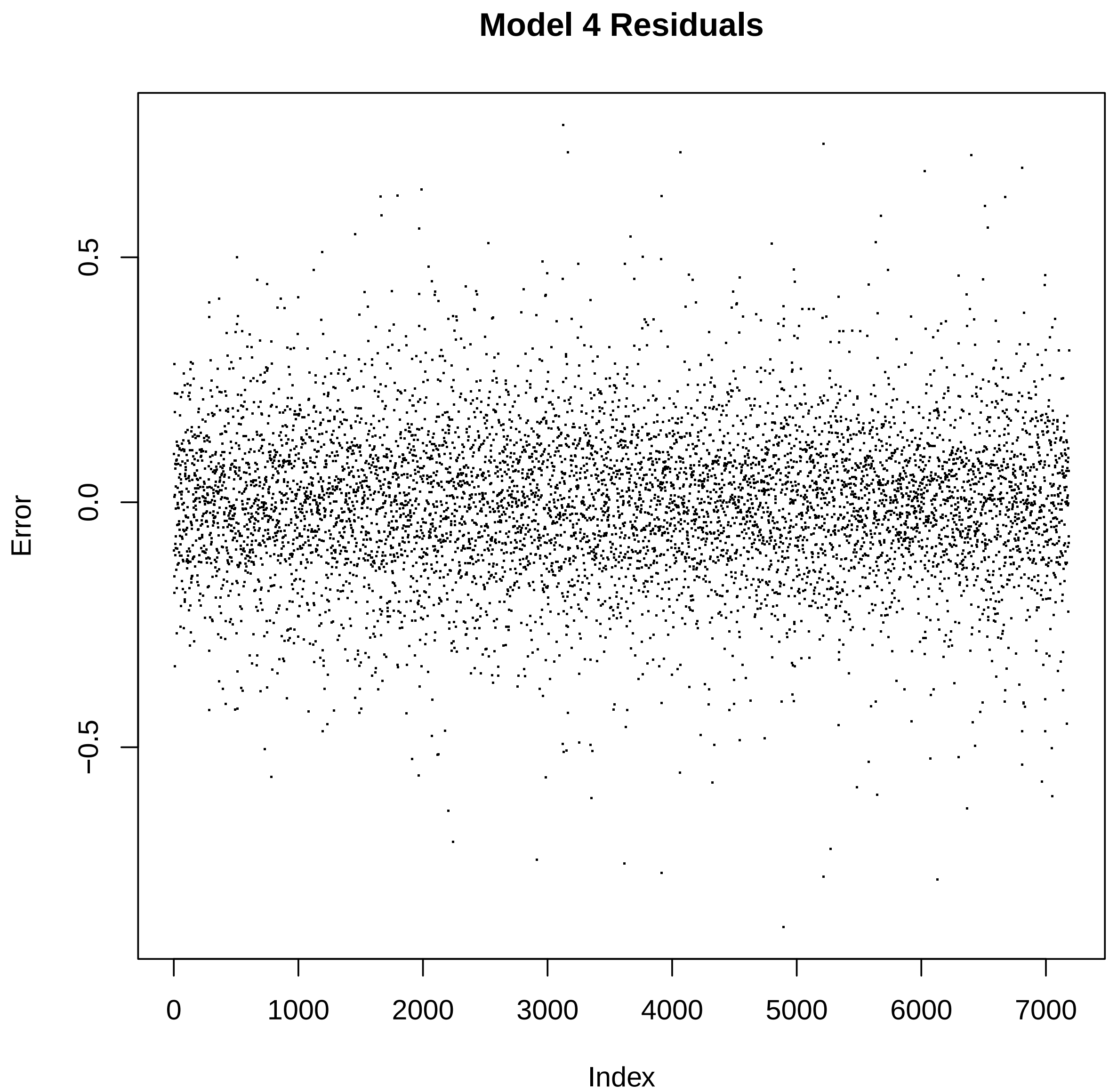

Model 3 Distribution
Frequency
0
500
1000
1500
2000
2500
3000
−1.0
−0.5
0.0
0.5
residuals(model_3)

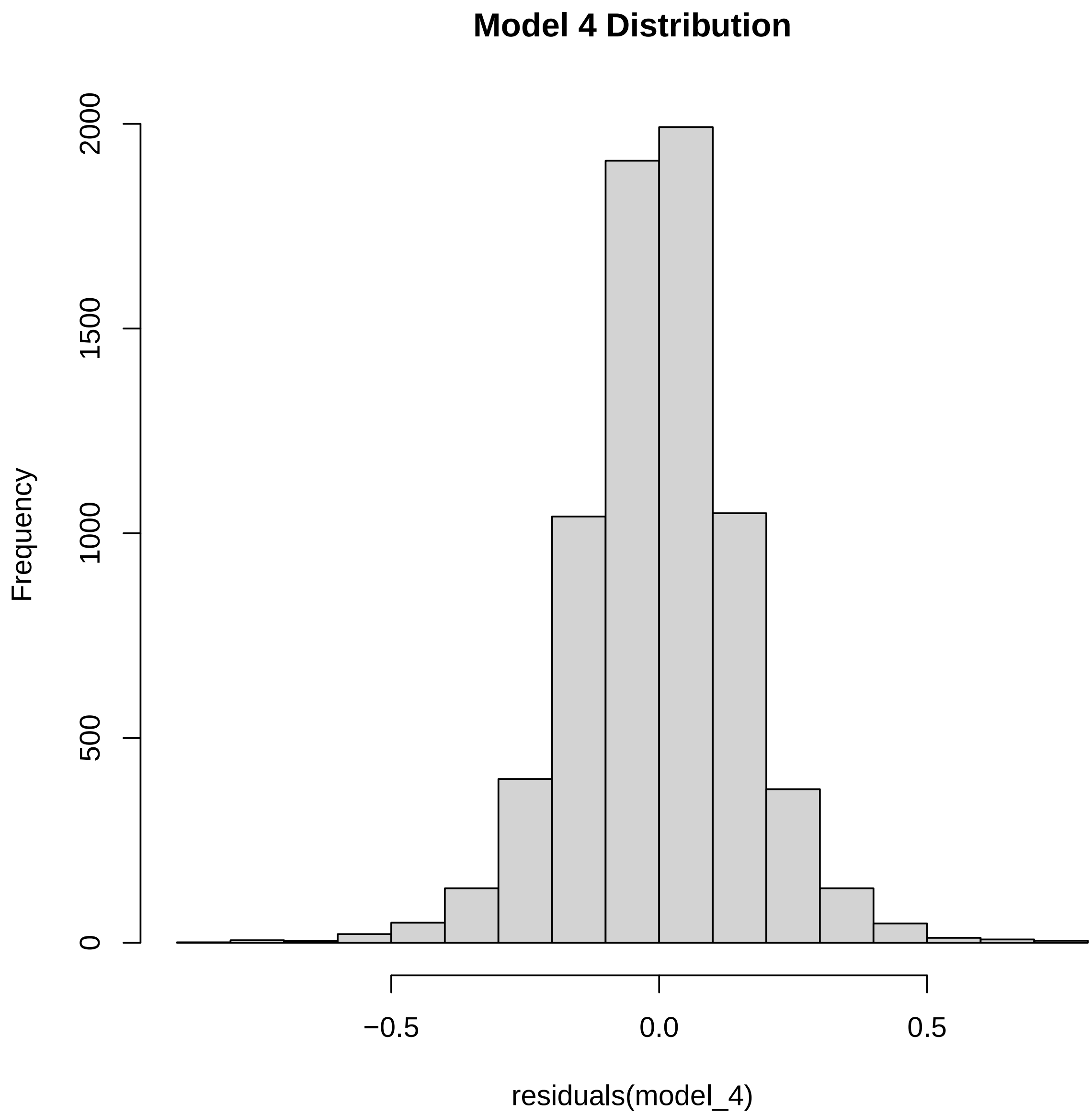


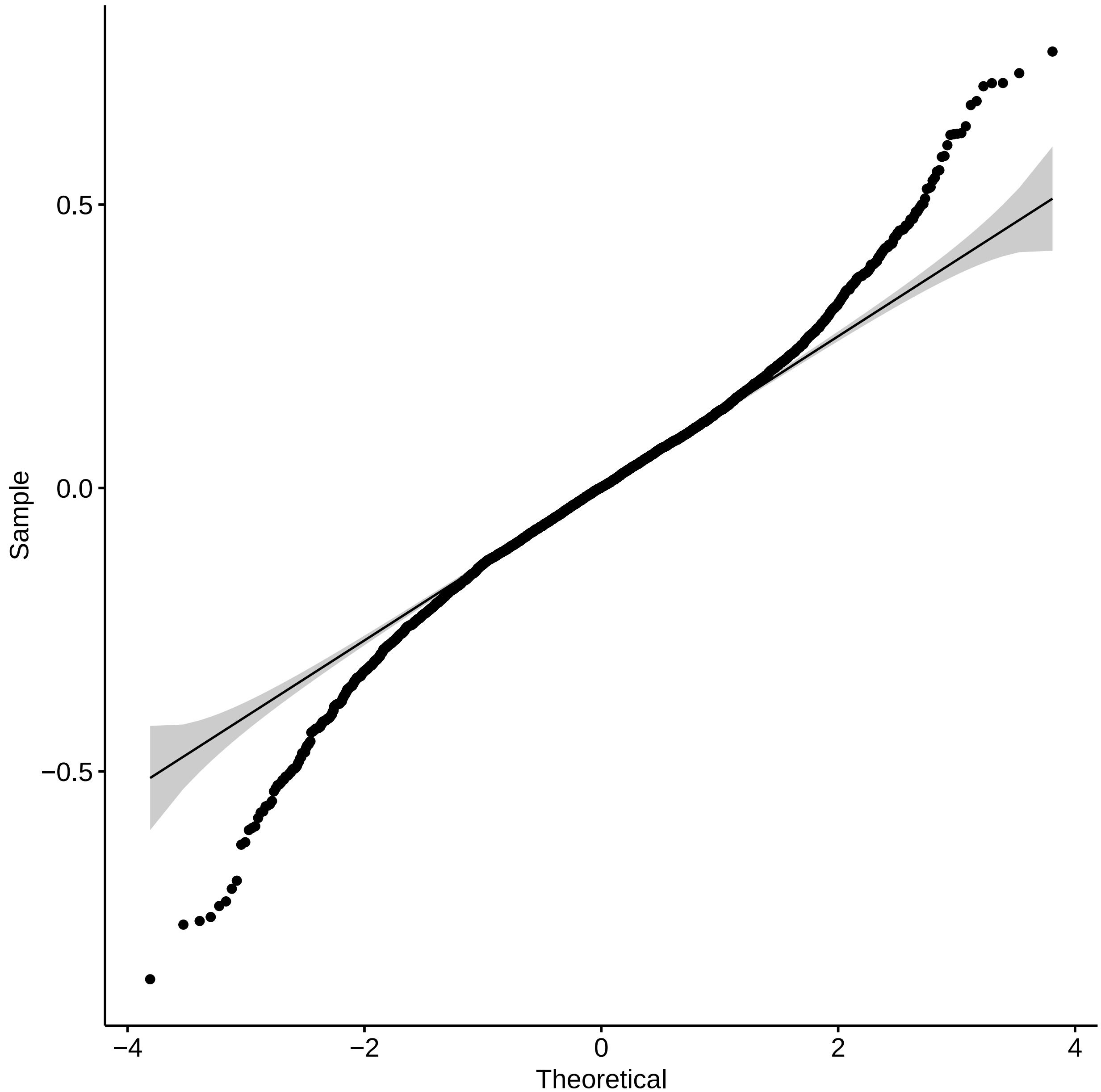

```
##
##  Box-Pierce test
##
## data:  residuals(model_3)
## X-squared = 76.079, df = 1, p-value < 2.2e-16
```



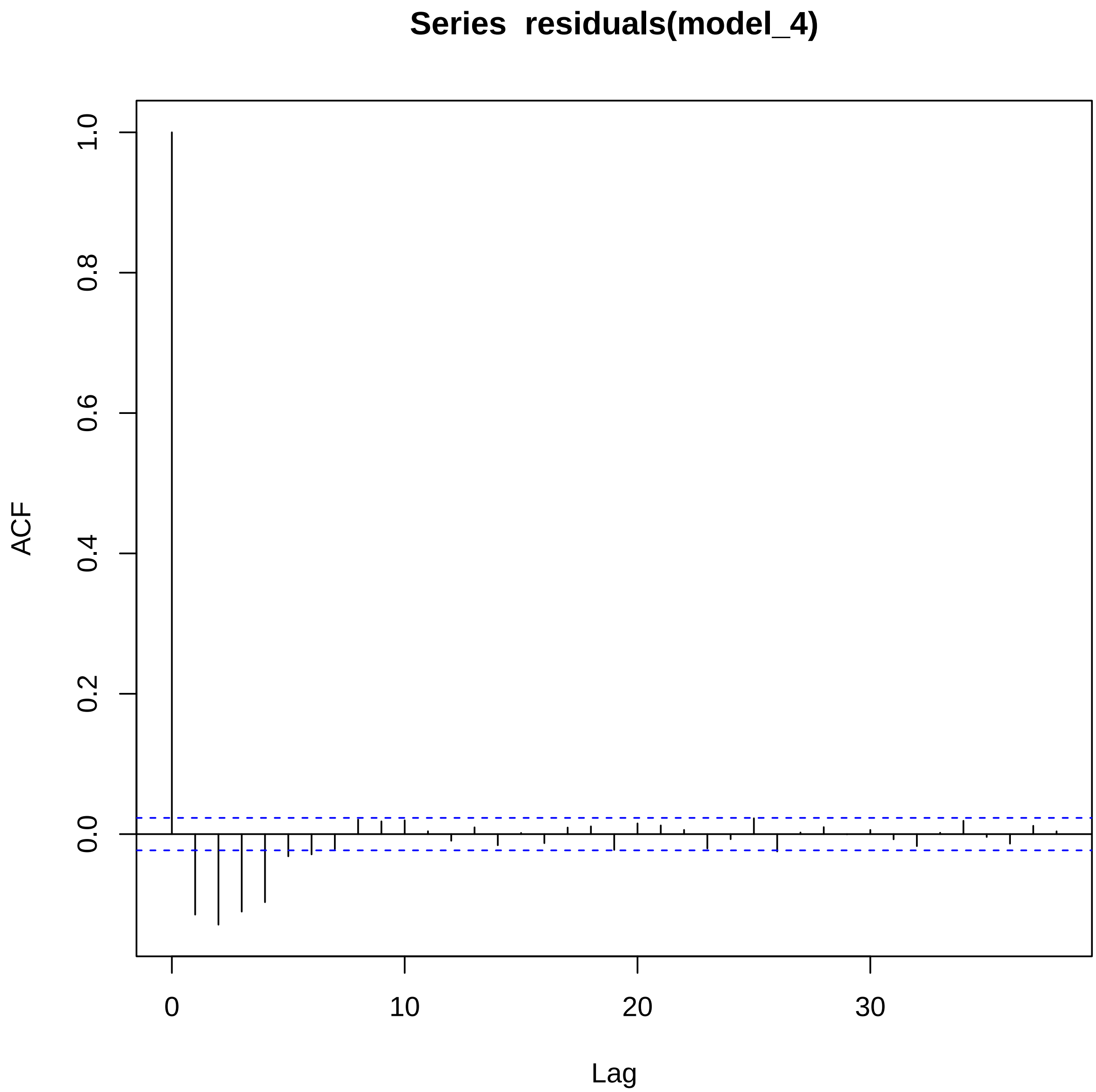
Model 4 Diagnostics


Model 4 Residuals
Error
0.5
0.0
−0.5
Index
0
1000
2000
3000
4000
5000
6000
7000


Model 4 Distribution
Frequency
0
500
1000
1500
2000
−0.5
0.0
0.5
residuals(model_4)




```
##
##  Box-Pierce test
##
## data:  residuals(model_4)
## X-squared = 94.508, df = 1, p-value < 2.2e-16
```



The scatter plots and histograms of the residuals appear to show that they are random and normally distributed (respectively) with mean 0. However the qq plots for all of the models, which show extremely heavy tails. Also, for all three models, the autocorrelation functions and box tests reveal that there is serial correlation between the residuals, meaning previous errors have an effect on future errors. All three tests yielded extremely significant results, which rejected the null hypothesis that there is no serial correlation in the residuals. All random effects for each of the models appear to be normally distributed centered around zero.

Model Comparisons To select an optimal model in answering the second research question, we will explore some model criterion.

```
##         df      AIC
## model_2 19 1396.069
```

```
## model_3 19 1420.615
## model_4 27 1472.878

## Analysis of Variance Table
##                         npar Sum Sq Mean Sq   F value
## year                       1  8.043   8.043  250.0080
## asthma                     1  0.155   0.155    4.8151
## smoke_ever                 1  1.092   1.092   33.9307
## mht                        1 66.280  66.280 2060.3490
## sex                        1  1.318   1.318   40.9725
## age_quartile               3 12.351   4.117  127.9755
## year:asthma                1  0.120   0.120    3.7310
## year:smoke_ever            1  0.070   0.070    2.1683
## asthma:smoke_ever          1  0.070   0.070    2.1644
## year:asthma:smoke_ever     1  0.022   0.022    0.6852
```

The first model, which used the binary variable smoke_ever, has the lowest AIC by a considerable margin, suggesting it has fit the data the best. The model is likely much more useful as far as making predictions for this reason.

Discussion

Results and Conclusions

Based on all the results, it can be concluded that the most significant variables affecting FEV are height, gender, whether or not the patient is asthmatic, and age. However, many of the variables tested with respect to time appear to be insignificant. That is, the models found no significant relationship between smoking status and FEV for asthmatics versus non-asthmatics., even when controlling for the other significant variables. *Asthmatics were found to have lower levels of lung function than non-asthmatics, but not steeper rates of FEV decline, regardless of smoking status.* This was shown by the significance of the parameter associated with variable "Asthmatic" in all four models, and the insignificance of the interaction parameters for asthma and smoking status - regardless of which smoking variable was utilized.

Future Work

It may be useful to adjust for weight and age simultaneously by including a categorical variable for BMI (body mass index). Another potential source of problems is the large number of missing values in the data set, more work could be done on handling this by either excluding all of the observations outright or imputing some reasonable values. Additionally, more model selection could be conducted in order to come up with a model with greater predictive utility. There is also an extreme amount of serial correlation in the residuals, which can dramatically reduce the value of the analyses. Possible means of addressing this could arise from time series methods, such as ARIMA modelling. The variable cncig is very problematic in the data set, as the vast majority of its values are either missing or zero. Imputing zero for the missing values skews the data to the right, and while this can be accounted for by converting the variable into categories by quantiles, said quantiles will still be very skewed. The lack of intuitive results is likely due to uneven groups. As shown before, there are *far* more non-asthmatics in the data set than asthmatics. Comparing groups of largely unequal sample sizes can easily lead to less reliable results as the data is dominated by people who do not have asthma. Perhaps a useful future study will be one that starts with an even number of participants in each group. That is, examining 100 people who are in each group.

References

https://www.bristol.ac.uk/cmm/media/research/ba-teaching-ebooks/pdf/Normality%20-%20Practical.pdf
https://cran.r-project.org/web/packages/lme4/vignettes/lmer.pdf
https://ladal.edu.au/regression.html#2__Mixed-Effects__Regression
https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5980468/
https://pubmed.ncbi.nlm.nih.gov/1644040/